

The MediaTek logo consists of the word "MEDIATEK" in white, uppercase, sans-serif font, centered within an orange parallelogram.

MT8668

Application Processor

Technical Brief

Version: 0.1
Release date: 2025-09-24

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Version History

Version	Date	Description
0.1	2025-09-24	Initial draft

Summary of Key Features

Please see Section 1.1.3 for more feature information.

Feature	Description
Security	
Chip-Level Features	Arm Trust Zone Security
Crypto Engine	The SSR, MediaTek's Unified Crypto Engine IP with Anti-SCA FIPS-140-3 ready for certification
Application Processor	
AP MCU	Arm® 2 little dual-core complex and 4 big cores
GPU	Arm® Mali GPU
SCP	MediaTek MRV55
Modem	
LTE Cat18, up to 4CC, NR sub6 3CC 5.14 Gbps	
Memory	
EMI	LPDDR5Maximum DRAM capacity: 24 GB Two channels, each having a 16-bit DQ bus
UFS	Supports UFS 3.1 2-lane and data rate up to 23.296 Gbps.
Multimedia	
Display	Display1: (1920x1080@60 + 1920x1080@60) Display2: (1920x1080@60 + 1920x1080@60) 2 x display with hardware CRC 2-layer telltale with 4 telltale checker
Camera	4 MIPI CSI-2 high-speed camera serial interfaces; max. 4 data lanes of D-DHY; max. 3 trios of C-PHY
Video	4K30 AV1/HEVC/AVC/VP9 video decoder 2K30 HEVC/AVC video encoder
Audio	
Modem Speech	
Connectivity	
USB, SPI, PWM, UART, GPIO, I2C/I3C, SIM, JTAG, MSDC (SD/SDIO 3.0), PCIe	
Wireless Connectivity	
Wi-Fi 6E 2x2, Bluetooth 5.4, GNSS L1+L5	
Process and Package	
4 nm process; 12.05 x 13.1 x 1.065mm, TFBGA	

MT8668 high-level block diagram

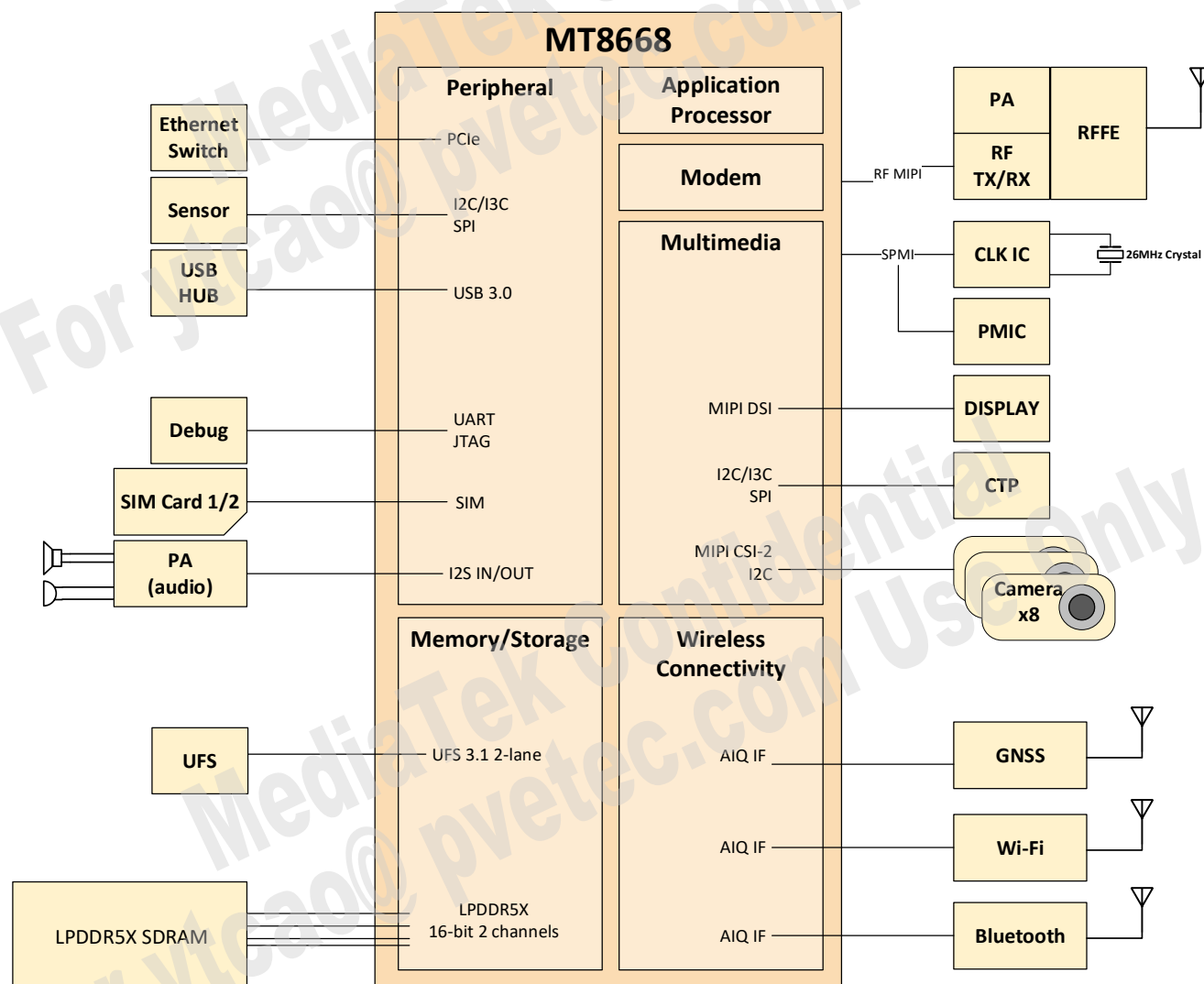


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1 Introduction

1.1 MT8668 Product Overview

MT8668 device is a highly integrated and scalable automotive application processor with rich multimedia features and AI capabilities. The chip integrates latest Arm CPU with 4 big cores and 2 little dual-core complex, along with a powerful multi-standard video codec. Furthermore, an extensive set of interfaces is integrated for camera modules, external audio components, UFS/SD cards, and external modules.

The Arm offers substantial computing power to support the latest open-source operating systems, as well as In-Vehicle-Infotainment (IVI) and Cockpit Domain Controller (CDC) applications. The Arm provides adequate computing power for dedicated tasks, such as software-based graphic rendering and real-time operating system in critical execution environments.

In addition, the SMMU with unique performance enhancements for hard real-time masters, integrated in MT8668, provides a straightforward hardware mechanism that minimizes the virtualization overhead of hypervisors. This is particularly beneficial when multiple operating systems coexist for various application domains, thus optimizing overall system performance.

The MDLA in MT8668 is capable to address the latest AI trends in achieving maximum effective performance in AI-multimedia, AI-gaming, AI-camera, and modern social video experiences. The MDLA is also aiming for ensuring the AI-enhanced technologies to work sustainably for the maximum duration in various conditions.

The multi-standard video codec in MT8668, supporting AV1, HEVC, AVC and more, offers advanced multimedia processing and multi-streaming audio and video capabilities.

The chip supports rich automotive camera features, such as around view monitoring, back view monitoring, automotive driving recoding, and driver monitoring. These features can be employed via the maximum 8x video stream inputs with MIPI-CSI2, or Ethernet AVB inputs. The display interface (MIPI DSI) allows for the support of up to 4 display panels, further enhancing its usability in various automotive applications.

1.1.1 Functional Block Diagram

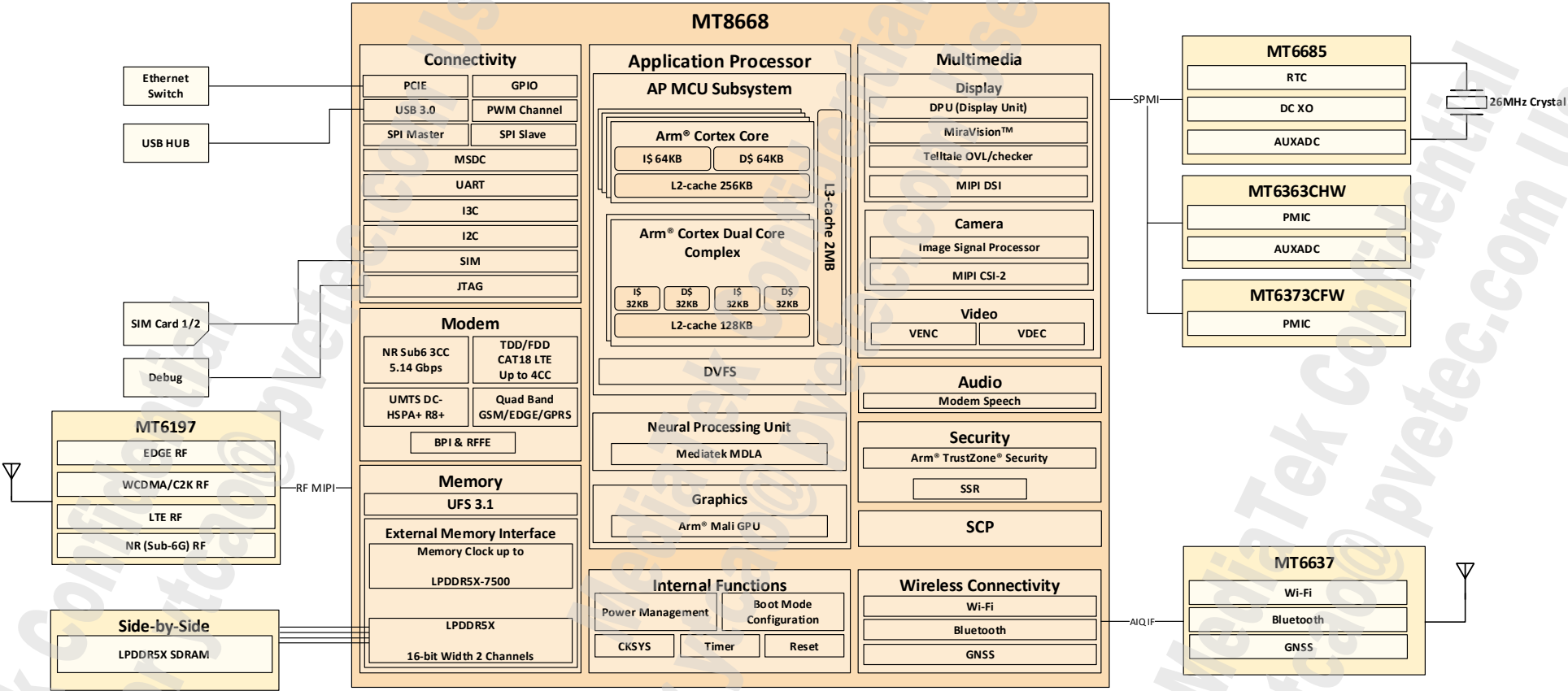


Figure 1-1. MT8668 functional block diagram

1.1.2 Power Domain

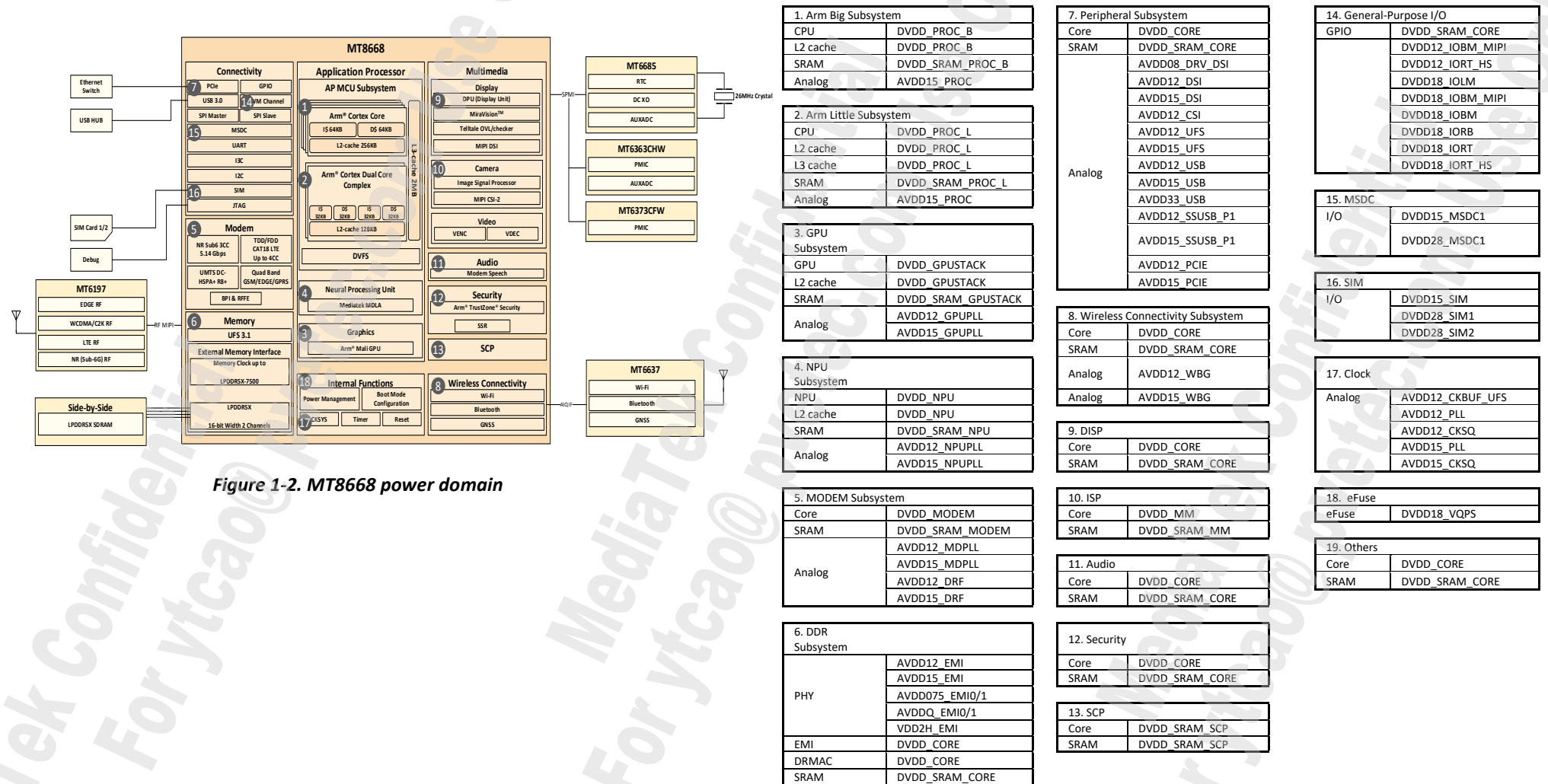


Figure 1-2. MT8668 power domain

1.1.3 Highlighted Features

Table 1-1. MT8668 highlighted features

MT8668	Capability
Security	
Chip-Level Features	• Arm® Trust Zone Security
Crypto Engine	• Scalable Security Root (SSR) – The SSR, the MediaTek's unified crypto engine IP, provides cryptographic functions for various products. – All crypto engines with anti-SCA – Pure hardware key protection – FIPS-140-3 ready for certification – Security IP/feature – Common crypto core ▪ SHA1-3 family/SM3 ▪ AES/TDES/DDES/SM4 crypto memory to memory engine – Pure hardware KDF ▪ HKDF/KBKDF/AMHK – NIST SP800-90C TRNG (HMAC_DRBG base) – MTK-RSA/ECC ▪ RSA 1024-8192 ▪ ECC NIST P192-P521 ▪ SM2 – PQC ▪ ML-DSA-44/65/87 ▪ ML-KEM-512/768/1024
Application Processor	
AP MCU	• Armv8.2 DSU with below configuration – 2 Arm® dual-core complex with 2 32 KB L1 I-Cache, 32 KB L1 D-Cache and 128 KB L2 Cache. – 4 Arm® cores with 64 KB L1 I-Cache, 64 KB L1 D-Cache and 256 KB L2 Cache. – Shared 2MB L3 Cache • DVFS technology supports for each core and DSU
GPU	• Arm Mali GPU • Per core capable of processing pixel rate 4M pixels per MHz, bi-linear texture rate 8M texels per MHz • OpenGL ES 3.2/3.1/3.0/2.0/1.1 • Vulkan 1.3 • OpenCL up to 3.0 • Supports raytracing. • Supports Variable Ray Shading (VRS).

Table 1-1. MT8668 highlighted features (continued)

MT8668	Capability
Neural Processing Unit	• NPU memory subsystem – DMA engine for data movement and data format conversion, image scaling and rotation • MediaTek Deep Learning Accelerator (MDLA) to support NN applications with high performance and high-power efficiency – Top performance: 8(A)x4(W) 12.5 TOPS, 8(A) x 8(W) 12.5 TOPS, 16(A) x 8(W) 6.2 TOPS, 16(A) x 16(W) 3.1 TOPS, FP16/BF16 3.1 TOPS – Supports layer based mixed precision – Simultaneous pipelined hardware function block (CONV/ACT/POOL/EWE/BILINEAR) – Supports Argmax – Enhancement of layer fusion to further reduce DRAM/TCM memory bandwidth – Enhancement of non-CONV OP performance – Supports Android NN asymmetric quantized data format • Supports weight decompression to reduce DRAM bandwidth
SCP	• MediaTek MRV55 • Built-in 1.25 MB Tightly Coupled Memory (TCM)
Modem	
NR sub6 3CC 5.14 Gbps	• Max. downlink bandwidth 3CC (220 M) (5.14 Gbps) • Max. uplink bandwidth 2CC (200 M) (2.5 Gbps) • Supports downlink 256 QAM/uplink 256 QAM. • Supports downlink MIMO 4x4/uplink MIMO 2x2.
LTE Cat18 Up to 4CC	• Downlink Cat18, 1.2 Gbps • Uplink Cat18, 211 Mbps • Supports downlink 256 QAM/uplink 256 QAM.
UMTS DC- HSPA+ R8+	• 3G modem supports most main features in 3GPP Release 7 and Release 8. • Uplink Cat. 7 (16 QAM) throughput up to 11.5 Mbps • Downlink Cat. 24 (64 QAM, dual-cell HSDPA) throughput up to 42.2 Mbps
BPI & RFFE	Baseband Parallel Interface (BPI) and MIPI RFFE interface with programmable driving strength multi-band
Quad Band GSM/EDGE/GPRS	GSM quad vocoders for Adaptive Multi-Rate (AMR), Enhanced Full Rate (EFR), Full Rate (FR) and Half Rate (HR)
Memory	
External Memory Interface (EMI)	• LPDDR5 • Maximum DRAM capacity: 24 GB • Two channels, each having a 16-bit DQ bus • Eight AXI subordinate interfaces • One AXI manager interface • Proprietary arbitration mechanism • Page-based memory protection • Customized QoS schemes • Resource monitors • Protocol checkers
UFS	Supports UFS 3.1 2-lane and data rate up to 23.296 Gbps.

Table 1-1. MT8668 highlighted features (continued)

MT8668	Capability
Multimedia	
Display	• Display1: (1920x1080@60 + 1920x1080@60) • Display2: (1920x1080@60 + 1920x1080@60) • MiraVision™ for picture quality enhancement • MIPI DSI transmitter with D-PHY 4+4 lanes, or C-PHY 3+3 trios • Embedded LCD gamma correction • True 10-bit colors • 8 overlay layers with per-pixel alpha channel and gamma table • 2 overlay layers which should be used on telltale • Spatial and temporal dithering • Color enhancement • Adaptive contrast enhancement • Image/video/graphic sharpness enhancement • Dynamic backlight scaling • Supports GPU decoder (AFBC). • Supports DSC compression.
Camera	• Multi-camera (8*YUV Sensor + 1*DP2CSI) – SVM [2.1 MP + 2.1 MP + 2.1 MP + 2.1 MP] @ 30 fps – RVM [2.1 MP] @ 60 fps – DMS [2.1 MP] @ 30 fps – OMS [2.1 MP] @ 30 fps – Dashcam (DVR) [2.1 MP] @ 30 fps – DP2CSI [2.1 MP] @ 30 fps • Support YUV 422 8/10/12 bit & RGB888 input • 4 MIPI CSI-2 high-speed camera serial interfaces; maximum of 4 data lanes of D-DHY; maximum of 3 trios of C-PHY
Video	• Video decoder (VDEC) – AV1 decoder: ▪ 4K2K@30 fps/100 Mbps, 8/10bits – HEVC decoder: ▪ Main profile 4K2K@30 fps/100 Mbps, 8bits ▪ Main 10 profile 4K2K@30 fps/100 Mbps, 10bits – VP9 decoder: ▪ Profile 0 4K2K@30 fps/100 Mbps, 8bits ▪ Profile 2 4K2K@30 fps/100 Mbps, 10bits – AVC decoder: ▪ Constrained baseline 4K2K@30 fps/100 Mbps, 8bits ▪ Main/high profile 4K2K@30 fps/100 Mbps, 8bits ▪ Progressive high 10 profile 4K2K@30 fps/100 Mbps, 10bits – HEIF decoder: ▪ Maximum resolution 16,383×16,383, 8/10bits • Video encoder (VENC) – H.264 encoder: 8-bit high profile 3,840×2,160@30 fps, 100 Mbps – HEVC encoder: 10-bit main profile 3,840×2,160@30 fps, 100 Mbps – HEIF encoder: Supported maximum resolution aligns image specification

Table 1-1. MT8668 highlighted features (continued)

MT8668	Capability
Audio	
Audio System	• Audio playing – Supports 8/11.025/12/16/22.05/24/32/44.1/48/96/192/384 kHz sampling rate playback. – Supports stereo audio playback. – Supports 3-channel playing with MT6368. • Audio recording – Supports 8/16/32/48/96/192 kHz sampling rate recording. – Supports recording stereo data. – Supports 3-channel recording with MT6369 or maximum 4-channel one-wire DMIC on the AP. • Speech – Supports 2 MICs. – Supports 8/16/32/48 kHz sampling rate recording. – Supports side tone filter.
Audio System	• I2S – Supports clock master and data input mode. – Supports clock master and data output mode. – Supports 16/24/32-bit stereo data. – Supports 8/11.025/12/16/22.05/24/32/44.1/48/88.2/96/176.4/192/384 kHz sampling rate in I2S output master mode. – Supports 8/11.025/12/16/22.05/24/32/44.1/48/88.2/96/176.4/192 kHz sampling rate in I2S input master mode. • Supports EIAJ/I2S format.
Modem Speech	• Speech codec • Cellular text telephone modem (CTM)
Connectivity	
USB	• USB 3.0 DRD
MSDC	• Port 1: SD/SDIO 3.0
UFS	• UFS 3.1 2-lane
SPI	• 8 SPI masters for external devices • Master mode only
PWM	• Max. 4 PWM channels (depending on system configuration and I/O usage)
UART	• 3 UARTs for debug and general application
I2C/I3C	• 13 I2Cs • 7 I3Cs • To control peripheral devices, e.g., CCM power, flash driver, audio amplifier, and OIS driver.
DISP_PWM	• 2 DISP_PWM • 4096 steps
GPIO	• The number of GPIO ports is 199.
SIM	• 1.8V/3V SIM cards using external level shifter • Supports output 20 MHz clock
JTAG	• JTAG for debugging and applications
PCIe	• PCIe Gen3 1-lane

Table 1-1. MT8668 highlighted features (continued)

MT8668	Capability
Wireless Connectivity	
Wi-Fi	• Wi-Fi 6E and backward compatible to legacy Wi-Fi spec (Wi-Fi 5/4) • Supports dual band (2.4 & 5~6 GHz band) • 2x2/1x1 with 80 MHz bandwidth, depending on companion connectivity chip capability • Integrated 2.4 GHz & 5GHz PA/LNA • Joint-TAS 2.0 • Antenna grouping • Security: – WPA personal/enterprise – WPA2 personal/enterprise – WPA3 personal/enterprise – WPS2.0 – WPI-SMS4 – WAPI (hardware)
Bluetooth	• BT5.4+ BLE Audio
GNSS	• GPS L1CA + L5 • BeiDou B1I + B2a • Glonass L1OF • Galileo E1 + E5a • QZSS L1CA + L1CB + L5 • NavIC N1 + N5
Process and Package	
Digital Die	4 nm process
Package	TFBGA
Package Size	12.05 x 13.1 x 1.065 mm
Companion Chips	See Section 8.2.

2 Terms and Abbreviations

2.1 Naming Convention

Table 2-1. MT8668 pin types

Abbreviation	Description
AI	Analog input
AO	Analog output
AIO	Analog bi-direction
DI	Digital input
DO	Digital output
DIO	Digital bi-direction
P	Power
G	Ground

2.2 Abbreviations

Abbreviation	Description
AE	Auto Exposure
AF	Auto Focus
AFBC	Arm Frame Buffer Compression
AMR	Adaptive Multi-Rate
AWB	Auto White Balance
BPI	Baseband Parallel Interface
BW	Bandwidth
CC	Closed Captioning; or Component Carrier
CDMA	Coda Division Multiple Access
CMD	Command
CMOS	Complementary Metal-Oxide Semiconductor
CSI	Camera Serial Interface
CSI-2	Camera Serial Interface 2
CTM	Cellular Text Telephone Modem
DB	Decibel
DBDC	Dual Band Dual Concurrent
DDR	Double Data Rate
DL	Downlink
DLA	Deep Learning Accelerator
DMA	Direct Memory Access
DMIC	Digital Microphone Interface Controller
DRAM	Dynamic Random-Access Memory

Abbreviation	Description
DSC	Display Stream Compression
DSI	Display Serial Interface
DSU	Data Service Unit
EC	Entropy encoding
EDGE	Enhanced Data Rates for GSM Evolution
EFR	Enhanced Full Rate
EMI	Electromagnetic Interference; External Memory Interface
ESD	Electrostatic Discharge
EV-DO	Evolution-Data Optimized (Evolution-Data Only)
EVS	Enhanced Voice Services
FDD	Frequency-Division Duplexing
FPU	Floating Point Unit
FR	Full Rate
GPIO	General-Purpose Input/Output
GPRS	General Packet Radio Service
GPS	Global Positioning System
GPU	Graphics Processing Unit
GSM	Global System for Mobile Communications
HEVC	High Efficiency Video Coding
HFIF	High Efficiency Image File
HR	Half Rate
HSUPA	High Speed Uplink Packet Access
HWROT	Hardware Root of Trust
IPD	Integrated Passive Devices
KBKDF	Key Based Key Derivation Function
LCD	Liquid Crystal Display
LCM	Liquid Crystal Module
LDO	Low Dropout
LSC	Land-Side Capacitors
MC	Motion Compensation
MCIA	MediaTek Confidentiality, Integrity, Anti-replay–Inline Cipher Engine
ME	Motion Estimation
MIMO	Multi-input Multi-Output
MIPI	Mobile Industry Processor Interface
MISO	Master in Slave Out
MLCC	Multilayer Ceramic Capacitors
MOSI	Master out Slave In
MSDC	Memory and SD Controller
NavIC	The NavIC or NAVigation with Indian Constellation is an autonomous regional satellite navigation system developed by Indian Space Research Organization (ISRO)
NB	Narrow Band

Abbreviation	Description
NN	Neural Network
NPU	Neural Processing Unit
NR	New Radio
PCIe	Peripheral Component Interconnect Express; PCI Express
PD	1. Pull Down; 2. Phase Detection
PLL	Phase-Locked Loop
PMIC	Power Management IC
PWM	Pulse-Width Modulation
QAM	Quadrature Amplitude Modulation
QHD	Quarter High Definition
QoS	Quality of Service
QSPI	Queued Serial Peripheral Interface
RC	Root Complex
RF	Radio Frequency
RF FE	Radio Frequency Front-End
ROM	Read-Only Memory
RTC	Real-Time Clock
SCK	Serial Clock
SCL	Serial Clock Line
SCP	System Companion Processor
SD	Secure Digital
SDA	Serial Data
SDIO	Secure Digital Input/Output
SLC	System Level Cache
SPI	Serial Peripheral Interface
SRAM	Static Random-Access Memory
SSR	Scalable Security Root
SWB	Super-Wideband
TCM	Tightly Coupled Memory
TDD	Time Division Duplex; Time Division Duplexing
TDM	Time Division Multiplex; Time Division Multiplexing
TQ	Transform and Quantization
UART	Universal Asynchronous Receiver and Transmitter
UFS	Universal Flash Storage
UL	Uplink
UMTS	Universal Mobile Telecommunications System
USB	Universal Serial Bus
VDC-M	VESA Display Stream Compression-M
VDEC	Video Decoder
VENC	Video Encoder

Abbreviation	Description
VPU	Vision Processing Unit
WB	Wideband
WBG	Wi-Fi/Bluetooth/GPS
WCG	Wide-Color-Gamut

3 Pin Information

The MT8668 pin information is distributed as shown below.

Section 3.1 The MT8668 pin map: [Table 3-1](#)

The function groups for the MT8668 pin map are categorized as follows.

	BT
	CSI
	DRF
	DSI
	EMI
	DVSS
	GNSS
	GPIO
	NC
	Power
	UFS
	USB
	XTAL_CLK

Section 3.2 Lists all the pin descriptions.

3.1 MT8668 Pin Map

Table 3-1. Pin map view

1084	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34			
A	NC_A1	NC_A2	EM1_D09		EM1_D03		EM1_D03		EM1_D01		EM1_D0		EM1_C4		EM1_C0		EM1_C3		EM1_C0		EM1_C5		EM1_D010		EM1_D0		EM1_D01		EM1_D04		EM1_D03		NC_A33	NC_A34	A		
B	NC_B1	DVSS	DVSS	EM1_D09	DVSS	EM1_D012	DVSS	EM1_D00		EM1_D02		EM1_C1		DVSS		EM1_C1		EM1_C3		DVSS		EM1_C4		EM1_D08	DVSS	EM1_D0		DVSS	EM1_D02	DVSS	EM1_D03	DVSS	EM1_D00	DVSS	NC_B34	B	
C	EM1_D05	EM1_W01		EM1_D0	DVSS	DVSS	EM1_D02		EM1_W01		EM1_D07		EM1_C6		NC		DVSS		NC		DVSS		EM1_D04		EM1_D02		EM1_D02		EM1_W01		EM1_RESET		DVSS		D		
D	EM1_D00	EM1_W01		EM1_D0		DVSS	EM1_D03		EM1_W01		EM1_D01		NC		NC		DVSS		NC		DVSS		EM1_D04		EM1_D02		EM1_D02		EM1_W01		EM1_RESET		DVSS		D		
E	EM1_D05	DVSS	DVSS	DVSS	DVSS	EM1_D01	DVSS	NC	NC	NC	NC		DVSS	DVSS	EM1_C1	EM1_C1					EM1_C1		EM1_D02		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	D		
F	AVDD15_WBG	WF1_OP	WF1_IN	WF1_IP	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	EM1_C5	EM1_C0					EM1_C6	EM1_C0			EM1_C0	EM1_C0															
G		WF1_QN	DVSS	DVSS	WFO_IN	WFO_IP	DVSS	DVSS	DVSS	DVSS	DVSS						DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS			
H	BT_IP		DVSS	DVSS	WFO_QN	WFO_QN	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS										DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		
J	BT_IN	BT_OP	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS										DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		
K		BT_QN	GL1_J	GL1_Q	GL1_J	DVSS	AVDD12_M_BG		DVSS	DVSS	DVSS	DVSS										DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		
L	DVSS	DVSS	GLS_Q		DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		
M	POE_CK_P	POE_CK_N	DVSS	DVSS	DVSS	POE_LNO_XP	POE_LNO_XP	DVSS	DVSS	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		
N	DVSS	DVSS	DVSS	DVSS	POE_LNO_XN	POE_LNO_XP	DVSS	DVSS	DVSS	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		
P		DVDD12_0	RT_HS																			DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS		
R	CONN_TO_P_CLK	CONN_TO_P_DATA	CONN_W_B_PTA	CONN_BT_DATA	CONN_BT_CLK	CONN_WF_CTRL4	CONN_WF_CTRL2	CONN_WF_CTRL3	CONN_WF_CTRL1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
T		ANT_SEL1	ANT_SEL2	ANT_SEL4	ANT_SEL3																	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
U	ANT_SEL2	DVSS	DVSS	DVSS	ANT_SEL1	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
V	DVSS	SCP_I2C_SD_A3	ANT_SEL0	CONN_HRS_T_B_2	CONN_HRS_T_B	GPS_L1_EL_NA_EN	GPS_L1_EL_NA_EN		DVSS	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
W	SCP_I2C_SDA2	SCP_I2C_SDA2	SCP_I2C_SCL3	DVDD12_0	SCP_I2C_SCL2	SCP_I2C_SCL2	SCP_I2C_SCL1	SCP_I2C_SDA1		DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
Y	SCP_I2C_SDA4	SCP_I2C_SDA4	SCP_I2C_SCL4	DVSS	DVDD12_0	I2SD0	I2SD1	I2SD2	I2SD3	DVDD12_0	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AA	SCP_SPI0_CK	SCP_SPI0_MO	SCP_SPI0_SB	SCP_SPI0_MI	DVSS	I2SD1	I2SD2	I2SD3	I2SD4	DVDD12_0	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AB	SCP_SPI1_CK	SCP_SPI1_MO	SCP_SPI1_MI	SCP_SPI1_SB	DVSS	I2SD1	I2SD2	I2SD3	I2SD4	DVDD12_0	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AC	DVSS	DVSS	SCP_SPI0_MO	SCP_SPI0_MI	SCP_SPI0_CK	DVSS	I2SD1	I2SD2	I2SD3	I2SD4	DVDD12_0	DVSS										DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AD	DVSS	DVSS	DVSS	DVSS	DVSS	SCP_SPI0_CK	SCP_SPI0_SB	SCP_SPI0_MI	SCP_SPI0_CK	DVSS	I2SD1	I2SD2	I2SD3	I2SD4	DVDD12_0	DVSS						DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AE	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AF	DVSS	DVSS	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AG	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AH	DVSS	DVSS	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AJ	DVSS	DVSS	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AK	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AL	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AM	DVSS	DVSS	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AN	DVSS	DVSS	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AP	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AR	DVSS	DVSS	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	CAM_CLK1	DVSS	DVSS											DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	
AT	NC_A1	NC_A2	EM1_D09		EM1_D03		EM1_D03		EM1_D01		EM1_D0		EM1_C4		EM1_C0		EM1_C3		EM1_C0		EM1_C5		EM1_D010		EM1_D0		EM1_D01		EM1_D04		EM1_D03		NC_A33	NC_A34	A		
AU	NC_A1	NC_A2	EM1_D09		EM1_D03		EM1_D03		EM1_D01		EM1_D0		EM1_C4		EM1_C0		EM1_C3		EM1_C0		EM1_C5		EM1_D010		EM1_D0		EM1_D01		EM1_D04		EM1_D03		NC_A33	NC_A34	A		

3.2 MT8668 Pin Descriptions

Table 3-2. MT8668 pin descriptions

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
System								
SYSRSTB	AR12	DI	System reset input	DVDD18	PD	I		
TESTMODE	AP12	DI	Test mode	DVDD18	PD	I		
AP_GOOD	AT12	DIO	When bootrom can normally jump into pre-loader execution (i.e., software can take over), bootrom makes AP_GOOD high before jumping	DVDD18	PD	OL	B:GPIO17	0
							O:AP_GOOD	1
							O:GPS_PPS	4
							O:AGPS_SYNC	6
X32K_IN	AP22	DIO	RTC 32K input	DVDD18	PD	I	B:GPIO187	0
							IO:RTC32K_CK	1
NFC_CLK_REQ	AE5	DIO	Clock request input pin for NFC	DVDD18	PD	I	B:GPIO52	0
							IO:SRCLKENAI0	1
							O:DISP_PWM1	3
							IO:SPM_JTAG_TRSTN_VLP	4
							IO:CONN_BGF_MCU_TRST_B	5
							O:SPMI_M_T_RIG_FLAG	6
SRCLKENAO	AM17	DIO	Output signal; control 26 Hz/Buck/LDO normal mode or sleep mode. (High: normal mode; low: sleep mode or low power mode)	DVDD18	PU	OH	B:GPIO184	0
							O:SRCLKENAO	1
WATCHDOG	AN17	DIO	Watchdog reset output	DVDD18	PD	OL	B:GPIO186	0

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:WATCHDOG	1
EINT_CHG_IRQB	AP16	DIO	External interrupt input for Charger	DVDD18	PD	I	B:GPIO0	0
							O:DISP_ERROR0	2
EINT1	W32	DIO	External interrupt input	DVDD18	PD	I	B:GPIO1	0
							O:TSFDC_EN	6
							I1:UFS_MPHY_SCL	7
EINT2	W33	DIO	External interrupt input	DVDD18	PD	I	B:GPIO2	0
							O:USB_DRVVBUS	1
							O:DISP_ERROR1	2
							O:TSFDC_TSS_ELO	6
							B1:UFS_MPHY_SDA	7
EINT3	W34	DIO	External interrupt input	DVDD18	PD	I	B:GPIO3	0
							I1:IDDIG	1
							O:TSFDC_TSS_EL1	6
EINT4	Y32	DIO	External interrupt input	DVDD18	PD	I	B:GPIO4	0
							I0:VBUSVALID	1
							O:CONN_TCXOENA_REQ	4
							O:TSFDC_TSS_EL2	6
EINT5	Y33	DIO	External interrupt input	DVDD18	PD	I	B:GPIO5	0
							B1:KPROW0	1
							O:GPS_PPS	6
EINT6	AT24	DIO	External interrupt input	DVDD18	PD	I	B:GPIO6	0
							O:DISP_PWM	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
EINT7	AU24	DIO	External interrupt input	DVDD18	PD	I	I1:UCTS2	4
							O:TSFDC_VCO_RST	6
							B:GPIO7	0
							O:DISP_PWM1	1
							IO:I2SIN_DMA0_DI3	2
EINT8	AU23	DIO	External interrupt input	DVDD18	PD	I	O:URTS2	4
							O:TSFDC_RCK_SELB	6
EINT9	AM23	DIO	External interrupt input	DVDD18	PD	I	B:GPIO8	0
							IO:DSI_TE1	1
							B:GPIO9	0
							O:LCM_RST1	1
EINT10	AR24	DIO	External interrupt input	DVDD18	PD	I	O:SPMI_P_TRIG_FLAG	6
							O:DBG_MON_B0	7
							B:GPIO10	0
							O:BPI_BUS10	1
							O:CONN_BPI_BUS10	2
EINT11	AP24	DIO	External interrupt input	DVDD18	PD	I	O:CONN_TCX_OENA_REQ	4
							O:DBG_MON_B1	7
							B:GPIO11	0
							O:BPI_BUS11	1
							O:CONN_BPI_BUS11_OLAT0	2
							O:HFRP_URTS1	5

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:DBG_MON_B2	7
EINT12	AN24	DIO	External interrupt input	DVDD18	PD	I	B:GPIO12	0
							O:BPI_BUS12	1
							O:CONN_BPI_BUS12_OLAT1	2
							O:ANT_SEL18	3
							O:CLKM0	4
							I1:HFRP_UCTS1	5
							O:DBG_MON_B3	7
EINT13	AM25	DIO	External interrupt input	DVDD18	PD	I	B:GPIO13	0
							O:BPI_BUS13	1
							O:CONN_BPI_BUS13_OLAT2	2
							O:ANT_SEL19	3
							O:CLKM1	4
							O:DBG_MON_B4	7
EINT14	AN25	DIO	External interrupt input	DVDD18	PD	I	B:GPIO14	0
							O:BPI_BUS14	1
							O:CONN_BPI_BUS14_OLAT3	2
							O:ANT_SEL20	3
							O:CLKM2	4
							O:DBG_MON_B5	7
EINT15	AM26	DIO	External interrupt input	DVDD18	PD	I	B:GPIO15	0
							O:BPI_BUS15	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:CONN_BPL_BUS15_OLAT4	2
							O:ANT_SEL21	3
							O:CLKM3	4
							O:DBG_MON_B6	7
SYS_OCP_UVL_O_INT	Y34	DIO	HWPT interrupt input	DVDD18	PU	I	B:GPIO198	0
							I0:MCU_PMIC_POC_I	1
X26M_IN	AA28	AI	26M clock input	AVDD12_CKSQ				
PMIC								
AUD_CLK_MO SI	AU20	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO188	0
							O:AUD_CLK_MOSI	1
							O:AUD_CLK_MOSI	3
AUD_SYNC_M OSI	AR20	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO189	0
							O:AUD_SYNC_MOSI	1
							I0:I2SIN_DMA0_BCK	4
							O:DBG_MON_B25	7
AUD_DAT_MO SI0	AR19	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO190	0
							O:AUD_DAT_MOSI0	1
							O:AUD_DAT_MOSI0	3
							O:DBG_MON_B26	7
AUD_DAT_MO SI1	AN20	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO191	0
							O:AUD_DAT_MOSI1	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:AUD_DAT_MOSI1	3
							O:DBG_MON_B27	7
AUD_DAT_MO SI2	AN21	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO191	0
							O:AUD_DAT_MOSI1	1
							O:AUD_DAT_MOSI1	3
							O:DBG_MON_B27	7
AUD_DAT_MO SI2	AN21	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO192	0
							O:AUD_DAT_MOSI2	1
							IO:I2SIN_DMA0_LRCK	4
							O:DBG_MON_B28	7
AUD_NLE_MO SIO	AT19	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO193	0
							O:AUD_NLE_MOSIO	1
							IO:AUD_SYNC_MISO	2
							IO:I2SIN_DMA0_DI	4
AUD_NLE_MO SI1	AT20	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO194	0
							O:AUD_NLE_MOSI1	1
							IO:AUD_CLK_MISO	2
							IO:I2SIN_DMA0_DI1	4
AUD_DAT_MIS O0	AU19	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO195	0
							IO:AUD_DAT_MISO0	1
							IO:VOW_DAT_MISO	2

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							IO:AUD_DAT_MISO0	3
							O:DBG_MON_B29	7
AUD_DAT_MISO1	AM20	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO196	0
							IO:AUD_DAT_MISO1	1
							IO:VOW_CLK_MISO	2
							IO:AUD_CLK_MISO	3
							IO:I2SIN_DMA0_DI2	4
							O:DBG_MON_B30	7
AUD_DAT_MISO2	AM21	DIO	PMIC audio input interface	DVDD18	PD	I	B:GPIO197	0
							IO:AUD_DAT_MISO2	1
							IO:AUD_DAT_MISO1	3
							IO:I2SIN_DMA0_DI3	4
							O:DBG_MON_B31	7
SPMI_M_SCL	AT27	DIO	PMIC SPMI control interface	DVDD18	NP	I	B:GPIO180	0
							B0:SPMI_M_SCL	1
SPMI_M_SDA	AU27	DIO	PMIC SPMI control interface	DVDD18	NP	I	B:GPIO181	0
							B0:SPMI_M_SDA	1
SPMI_P_SCL	AN28	DIO	PMIC SPMI control interface	DVDD18	NP	I	B:GPIO182	0
							B0:SPMI_P_SCL	1
SPMI_P_SDA	AP28	DIO	PMIC SPMI control interface	DVDD18	NP	I	B:GPIO183	0
							B0:SPMI_P_SDA	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
SIM								
SIM1_SCLK	AM15	DIO	SIM1 clock, PMIC interface	DVDD28_SIM1	PD	I	B:GPIO174	0
							O:MD1_SIM1_SCLK	1
							IO:TSFDC_26M	6
SIM1_SRST	AM16	DIO	SIM1 data, PMIC interface	DVDD28_SIM1	PD	I	B:GPIO175	0
							O:MD1_SIM1_SRST	1
							I1:HFRP_JTAG0_TCK	2
							I1:MCUPM_JTAG_TCK	3
							IO:APU_JTAG_TCK	4
							IO:CONN_DSP_L5_JCK	5
							O:TSFDC_SDO	6
SIM1_SIO	AN16	DIO	SIM1 data, PMIC interface	DVDD28_SIM1	PD	I	B:GPIO176	0
							B1:MD1_SIM1_SIO	1
							IO:HFRP_JTAG0_TRSTN	2
							IO:MCUPM_JTAG_TRSTN	3
							IO:APU_JTAG_TRST	4
							O:CONN_DSP_L5_JINTP	5
							O:TSFDC_FOUT	6
SIM2_SCLK	AN15	DIO	SIM2 clock, PMIC interface	DVDD28_SIM2	PD	I	B:GPIO177	0
							O:MD1_SIM2_SCLK	1
							I1:HFRP_JTAG0_TDI	2

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							I1:MCUPM_JTAG_TDI	3
							I1:APU_JTAG_TDI	4
							I0:CONN_DSP_L5_JDI	5
							I0:TSFDC_SCK	6
SIM2_SRST	AM14	DIO	SIM2 data, PMIC interface	DVDD28_SIM2	PD	I	B:GPIO178	0
							O:MD1_SIM2_SRST	1
							I1:HFRP_JTAG_0_TMS	2
							I1:MCUPM_JTAG_TMS	3
							B1:APU_JTAG_TMS	4
							I1:CONN_DSP_L5_JMS	5
							I0:TSFDC_SDI	6
SIM2_SIO	AN14	DIO	SIM2 data, PMIC interface	DVDD28_SIM2	PD	I	B:GPIO179	0
							B1:MD1_SIM2_SIO	1
							O:HFRP_JTAG_0_TDO	2
							O:MCUPM_JTAG_TDO	3
							O:APU_JTAG_TDO	4
							O:CONN_DSP_L5_JDO	5
							I0:TSFDC_SCF	6
INT_SIM1	AN23	DIO	SIM1 interrupt	DVDD18	PD	I	B:GPIO119	0
							I0:MD_INT1_C2K_UIM0_HOT_PLUG	1
							O:SRCLKENA1	5

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
INT_SIM2	AM24	DIO	SIM2 interrupt	DVDD18	PD	I	B:GPIO120	0
							I0:MD_INT2_C2K_UIM1_HOT_PLUG	1
							B0:TP_GPIO15_AO	4
							O:SRCLKENA2	5
I2S								
I2S1_BCK	AR21	DIO	I2S interface	DVDD18	PD	I	B:GPIO70	0
							O:I2SIN1_BCK	1
							I0:I2SIN_DMA0_BCK	2
							I0:MFG_EB_JTAG_TCK	4
							I1:SCP_JTAG0_TCK_VCORE	6
							I0:CONN_WF_MCU_TCK	7
I2S1_LRCK	AU22	DIO	I2S interface	DVDD18	PD	I	B:GPIO71	0
							O:I2SIN1_LRCK	1
							I0:I2SIN_DMA0_LRCK	2
							I0:MFG_EB_JTAG_TMS	4
							I1:SCP_JTAG0_TMS_VCORE	6
							I1:CONN_WF_MCU_TMS	7
I2S1_DI	AU21	DIO	I2S interface	DVDD18	PD	I	B:GPIO72	0
							I0:I2SIN1_DI	1
							I0:I2SIN_DMA0_DI	2
							I0:MFG_EB_JTAG_TDI	4

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							I1:SCP_JTAG0_TDI_VCORE	6
							I0:CONN_WF_MCU_TDI	7
I2S1_DO	AT21	DIO	I2S interface	DVDD18	PD	I	B:GPIO73	0
							O:I2SOUT1_DO	1
							I0:I2SIN_DMA0_DI1	2
							O:MFG_EB_JTAG_TDO	4
							O:SCP_JTAG0_TDO_VCORE	6
							O:CONN_WF_MCU_TDO	7
I2S1_MCK	AT22	DIO	I2S interface	DVDD18	PD	I	B:GPIO74	0
							O:I2SIN1_MCK	1
							I0:I2SIN_DMA0_DI2	2
							I0:MFG_EB_JTAG_TRSTN	4
							I0:SCP_JTAG0_TRSTN_VCORE	6
							I0:CONN_WF_MCU_TRST_B	7
I2S2_BCK	Y7	DIO	I2S interface	DVDD18	PD	I	B:GPIO75	0
							O:I2SIN2_BCK	1
							I1:SCP_JTAG0_TCK_VLP	5
							I1:SSPM_JTAG_TCK_VLP	6
							I0:IO_JTAG_TCK	7
I2S2_LRCK	AA8	DIO	I2S interface	DVDD18	PD	I	B:GPIO76	0

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:I2SIN2_LRC K	1
							I1:SCP_JTAG0 _TMS_VLP	5
							I1:SSPM_JTA G_TMS_VLP	6
							B1:IO_JTAG_T MS	7
I2S2_DI	Y8	DIO	I2S interface	DVDD18	PD	I	B:GPIO77	0
							IO:I2SIN2_DI	1
							I1:SCP_JTAG0 _TDI_VLP	5
							I1:SSPM_JTA G_TDI_VLP	6
							I1:IO_JTAG_T DI	7
I2S2_DO	Y6	DIO	I2S interface	DVDD18	PD	I	B:GPIO78	0
							O:I2SOUT2_D O	1
							O:SCP_JTAG0 _TDO_VLP	5
							O:SSPM_JTAG _TDO_VLP	6
							O:IO_JTAG_T DO	7
I2S2_MCK	AA7	DIO	I2S interface	DVDD18	PD	I	B:GPIO79	0
							O:I2SIN2_MC K	1
							IO:SCP_JTAG0 _TRSTN_VLP	5
							IO:SSPM_JTA G_TRSTN_VLP	6
							IO:IO_JTAG_T RSTN	7
Liquid crystal display (LCD)								
DSI_TE	AR22	DIO	Parallel display interface tearing effect	DVDD18	PD	I	B:GPIO59	0

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							I0:DSI_TE	1
							O:DBG_MON_A28	7
LCM_RST	AT23	DIO	Parallel display interface reset signal	DVDD18	PD	I	B:GPIO60	0
							O:LCM_RST	1
							O:DBG_MON_A29	7
Pulse width modulation (PWM)								
PWM_OUT	AR23	DIO	Display PWM output	DVDD18	PD	I	B:GPIO61	0
							O:DISP_PWM	1
							O:PWM_0	3
Keypad interface								
KPCOLO	AR16	DIO	Keypad column 0	DVDD18	PU	I	B:GPIO16	0
							B1:KPCOLO_VLP	1
Serial peripheral interface (SPI)								
SPI4_CK	R29	DIO	SPI4 clock	DVDD18	PD	I	B:GPIO32	0
							O:SPI4_CLK	1
							O:I2SINO_BCK	2
							O:CLKM0	4
							O:CMFLASH0	5
							I1:UCTS3	6
							O:DBG_MON_A12	7
SPI4_CSB	R30	DIO	SPI4 chip select	DVDD18	PD	I	B:GPIO33	0
							O:SPI4_CSB	1
							O:I2SINO_LRC K	2
							O:CLKM1	4
							O:CMFLASH1	5

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
SPI4_MO	R32	DIO	SPI4 data out	DVDD18	PD	I	O:URTS3	6
							O:DBG_MON_A13	7
							B:GPIO34	0
							B0:SPI4_MO	1
							O:I2SIN0_MCK	2
							O:CLKM2	4
							O:CMFLASH2	5
SPI4_MI	R33	DIO	SPI4 data in	DVDD18	PD	I	O:UTXD3	6
							O:DBG_MON_A14	7
							B:GPIO35	0
							B0:SPI4_MI	1
							IO:I2SIN0_DI	2
							O:CLKM3	4
							O:CMFLASH3	5
SPI5_CK	P30	DIO	SPI5 clock	DVDD18	PD	I	I1:URXD3	6
							O:DBG_MON_A15	7
							B:GPIO28	0
							O:SPI5_CLK	1
							B0:FMI2SIN_BCK	2
							O:CLKM0	4
							IO:DAP_SONIC_SWCK	5
SPI5_CSB	N29	DIO	SPI5 chip select	DVDD18	PD	I	O:MD_UTXD1	6
							O:DBG_MON_A8	7
							B:GPIO29	0

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:SPI5_CSB	1
							B0:FMI2SIN_L RCK	2
							O:CLKM1	4
							B1:DAP_SONIC_SWDC	5
							I1:MD_URXD1	6
							O:DBG_MON_A9	7
SPI5_MO	P28	DIO	SPI5 data out	DVDD18	PD	I	B:GPIO30	0
							B0:SPI5_MO	1
							O:FMI2SIN_MCK	2
							O:CLKM2	4
							I0:DAP_MD32_SWCK	5
							O:DBG_MON_A10	7
SPI5_MI	P29	DIO	SPI5 data in	DVDD18	PD	I	B:GPIO31	0
							B0:SPI5_MI	1
							I0:FMI2SIN_DI	2
							O:CLKM3	4
							B1:DAP_MD32_SWDC	5
							O:DBG_MON_A11	7
SPI6_CLK	P33	DIO	SPI6 clock	DVDD18	PD	I	B:GPIO24	0
							O:SPI6_CLK	1
							I0:SPIS1_CLK	2
							B0:TP_GPIO0_AO	4

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:DBG_MON_A4	7
SPI6_CSB	P34	DIO	SPI6 chip select	DVDD18	PD	I	B:GPIO25	0
							O:SPI6_CSB	1
							I1:SPIS1_CSB	2
							B0:TP_GPIO1_AO	4
							O:DBG_MON_A5	7
SPI6_MO	P31	DIO	SPI6 data out	DVDD18	PD	I	B:GPIO26	0
							B0:SPI6_MO	1
							B0:SPIS1_MO SI	2
							O:DMIC_CLK	6
							O:SCP_DMIC_CLK	4
SPI6_MI	P32	DIO	SPI6 data in	DVDD18	PD	I	O:DBG_MON_A6	7
							B:GPIO27	0
							B0:SPI6_MI	1
							B0:SPIS1_MIS O	2
							IO:DMIC_DAT	3
							IO:SCP_DMIC_DAT	4
SPI7_CK	M33	DIO	SPI7 clock	DVDD18	PD	I	O:DBG_MON_A7	7
							B:GPIO20	0
							O:SPI7_CLK	1
							IO:SPIS0_CLK	2
							O:MD_URTS0	5
							O:UTXD4	6

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:DBG_MON_A0	7
SPI7_CSB	M32	DIO	SPI7 chip select	DVDD18	PD	I	B:GPIO21	0
							O:SPI7_CSB	1
							I1:SPIS0_CSB	2
							I1:MD_UCTS0	5
							I1:URXD4	6
							O:DBG_MON_A1	7
SPI7_MO	N33	DIO	SPI7 data out	DVDD18	PD	I	B:GPIO22	0
							B0:SPI7_MO	1
							B0:SPIS0_MO SI	2
							O:MD_URTS1	5
							O:UTXD5	6
							O:DBG_MON_A2	7
SPI7_MI	N32	DIO	SPI7 data in	DVDD18	PD	I	B:GPIO23	0
							B0:SPI7_MI	1
							B0:SPIS0_MISO	2
							I1:MD_UCTS1	5
							I1:URXD5	6
							O:DBG_MON_A3	7
UART								
UTXD0	AM12	DIO	UR0-TX	DVDD18	PU	OH	B:GPIO167	0
							O:UTXD0	1
URXD0	AN12	DIO	UR0-RX	DVDD18	PU	I	B:GPIO168	0
							I1:URXD0	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
UTXD1	AN11	DIO	UR1-TX	DVDD18	PD	I	B:GPIO18	0
							O:UTXD1	1
							O:TP_UTXD1_VCORE	3
							O:HFRP_UTXD1	5
							O:MD_UTXD0	6
							O:CONN_WIFI_TXD	7
URXD1	AM11	DIO	UR1-RX	DVDD18	PD	I	B:GPIO19	0
							I1:URXD1	1
							I1:TP_URXD1_VCORE	3
							I1:HFRP_URXD1	5
							I1:MD_URXD0	6
							RF-interface	
BPI_D_BUS0	AN27	DIO	BPI_BUS0	DVDD18	PD	I	B:GPIO147	0
							O:BPI_BUS0	1
							O:ANT_SEL8	3
							O:CLKM0	4
							O:DBG_MON_B15	7
BPI_D_BUS1	AP27	DIO	BPI_BUS1	DVDD18	PD	I	B:GPIO148	0
							O:BPI_BUS1	1
							O:ANT_SEL9	3
							O:CLKM1	4
							O:DBG_MON_B16	7
BPI_D_BUS2	AN26	DIO	BPI_BUS2	DVDD18	PD	I	B:GPIO149	0
							O:BPI_BUS2	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:ANT_SEL10	3
							O:CLKM2	4
							O:DBG_MON_B17	7
BPI_D_BUS3	AP26	DIO	BPI_BUS3	DVDD18	PD	I	B:GPIO150	0
							O:BPI_BUS3	1
							O:ANT_SEL11	3
							O:CLKM3	4
							I0:MD_GPS_B_LANK_EXT	6
							O:DBG_MON_B18	7
BPI_D_BUS4	AR26	DIO	BPI_BUS4	DVDD18	PD	I	B:GPIO151	0
							O:BPI_BUS4	1
							O:ANT_SEL12	3
							I1:MD32_0_RXD	6
BPI_D_BUS5	AU26	DIO	BPI_BUS5	DVDD18	PD	I	O:DBG_MON_B19	7
							B:GPIO152	0
							O:BPI_BUS5	1
							O:ANT_SEL13	3
BPI_D_BUS6	AT26	DIO	BPI_BUS6	DVDD18	PD	I	O:MD32_0_TXD	6
							O:DBG_MON_B20	7
							B:GPIO153	0
							O:BPI_BUS6	1
							O:CONN_BPI_BUS6	2
							O:ANT_SEL14	3

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							B0:MD32_0_GPIO0	6
							O:DBG_MON_B21	7
BPI_D_BUS7	AR25	DIO	BPI_BUS7	DVDD18	PD	I	B:GPIO154	0
							O:BPI_BUS7	1
							O:CONN_BPI_BUS7	2
							O:ANT_SEL15	3
							I1:MD32_1_RXD	6
							O:DBG_MON_B22	7
BPI_D_BUS8	AU25	DIO	BPI_BUS8	DVDD18	PD	I	B:GPIO155	0
							O:BPI_BUS8	1
							O:CONN_BPI_BUS8	2
							O:ANT_SEL16	3
							O:MD32_1_TXD	6
							O:DBG_MON_B23	7
BPI_D_BUS9	AT25	DIO	BPI_BUS9	DVDD18	PD	I	B:GPIO156	0
							O:BPI_BUS9	1
							O:CONN_BPI_BUS9	2
							O:ANT_SEL17	3
							B0:MD32_1_GPIO0	6
							O:DBG_MON_B24	7
MD_UCNT	AU32	DIO	BPI control	DVDD18	PD	I	B:GPIO157	0
							I0:MD_UCNT_A_TGL	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
DIGRF_IRQ	AR27	DIO	BPI control	DVDD18	PD	I	B:GPIO158	0
							IO:DIGRF_IRQ	1
RF-MIPI-interface								
MIPI_M_SCLK	AU30	DIO	MIPI_M CLK	DVDD18	PD	I	B:GPIO159	0
							O:MIPI_M_SC LK	1
MIPI_M_SDAT A	AU31	DIO	MIPI_M DATA	DVDD18	PD	I	B:GPIO160	0
							B0:MIPI_M_S DATA	1
MIPIO_D_SCLK	AT29	DIO	MIPI_0 CLK	DVDD18	PD	I	B:GPIO161	0
							O:MIPIO_D_S CLK	1
							O:CONN_MIPI 0_SCLK	2
							O:BPI_BUS16	3
							O:CONN_BPI_ BUS16_OLAT 5	4
							O:MD_GPS_L 1_BLANK	6
MIPIO_D_SDAT A	AT30	DIO	MIPI_0 DATA	DVDD18	PD	I	B:GPIO162	0
							B0:MIPIO_D_S DATA	1
							B0:CONN_MI PIO_SDATA	2
							O:BPI_BUS17	3
							O:CONN_BPI_ BUS17_ANT0	4
							O:MD_GPS_L 5_BLANK	6
MIPI1_D_SCLK	AT28	DIO	MIPI_1 CLK	DVDD18	PD	I	B:GPIO163	0
							O:MIPI1_D_S CLK	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:CONN_MIPI1_SCLK	2
							O:BPI_BUS18	3
							O:CONN_BPI_BUS18_ANT1	4
							O:MIPI3_D_SCLK	5
MIPI1_D_SDAT A	AR28	DIO	MIPI_1 DATA	DVDD18	PD	I	B:GPIO164	0
							B0:MIPI1_D_SDATA	1
							B0:CONN_MIPI1_SDATA	2
							O:BPI_BUS19	3
							O:CONN_BPI_BUS19_ANT2	4
							B0:MIPI3_D_SDATA	5
MIPI2_D_SCLK	AU28	DIO	MIPI_2 CLK	DVDD18	PD	I	IO:MD_GPS_B LANK_EXT	6
							B:GPIO165	0
							O:MIPI2_D_SCLK	1
							O:BPI_BUS20	3
							O:CONN_BPI_BUS20_ANT3	4
MIPI2_D_SDAT A	AU29	DIO	MIPI_2 DATA	DVDD18	PD	I	O:MIPI4_D_SCLK	5
							B:GPIO166	0
							B0:MIPI2_D_SDATA	1
							O:BPI_BUS21	3
							O:CONN_BPI_BUS21_ANT4	4
							B0:MIPI4_D_SDATA	5

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
MSDC1								
MSDC1_CLK	AU15	DIO	SD/SDIO interface	DVDD28_MSDC1	PD	I	B:GPIO53	0
							B0:MSDC1_CLK	1
							O:MD1_SIM2_SCLK	2
							IO:CONN_DSP_JCK	3
							IO:UDI_TCK	4
							IO:IPU_JTAG_TCK	5
							I1:SSPM_JTAG_TCK_VCORE	6
MSDC1_CMD	AP15	DIO	SD/SDIO interface	DVDD28_MSDC1	PD	I	B:GPIO54	0
							B1:MSDC1_CMD	1
							B1:CONN_BGF_MCU_AICE_TMSC	2
							I1:CONN_DSP_JMS	3
							IO:UDI_TMS	4
							IO:IPU_JTAG_TMS	5
							I1:SSPM_JTAG_TMS_VCORE	6
MSDC1_DAT0	AR15	DIO	SD/SDIO interface	DVDD28_MSDC1	PD	I	B:GPIO55	0
							B1:MSDC1_DAT0	1
							O:MD1_SIM2_SRST	2
							IO:CONN_DSP_JDI	3
							IO:UDI_TDI_0	4

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							IO:IPU_JTAG_TDI	5
							I1:SSPM_JTAG_TDI_VCORE	6
MSDC1_DAT1	AT15	DIO	SD/SDIO interface	DVDD28_MSDC1	PD	I	B:GPIO56	0
							B1:MSDC1_DAT1	1
							B1:MD1_SIM2_SIO	2
							O:CONN_DSP_JDO	3
							O:UDI_TDO_0	4
							O:IPU_JTAG_TDO	5
							O:SSPM_JTAG_TDO_VCORE	6
							IO:CONN_WF_MCU_AICE_TCKC	7
MSDC1_DAT2	AU16	DIO	SD/SDIO interface	DVDD28_MSDC1	PD	I	B:GPIO57	0
							B1:MSDC1_DAT2	1
							IO:CONN_BGF_MCU_AICE_TCKC	2
							O:CONN_DSP_JINTP	3
							IO:UDI_NTRST	4
							IO:IPU_JTAG_TRST	5
							IO:SSPM_JTAG_TRSTN_VCORE	6
							B:GPIO58	0
MSDC1_DAT3	AT16	DIO	SD/SDIO interface	DVDD28_MSDC1	PD	I	B1:MSDC1_DAT3	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:CONN_TCX OENA_REQ	4
							O:AGPS_SYNC	6
							B1:CONN_WF _MCU_AICE_ TMSC	7
Camera(CAM)								
CAM_RST0	AM8	DIO	Reset control to 0th sensor	DVDD18	PD	I	B:GPIO62	0
							O:BPI_BUS22	1
							I1:UCTS2	3
							I0:UDI_TDI_1	4
CAM_RST1	AN7	DIO	Reset control to 1st sensor	DVDD18	PD	I	B:GPIO63	0
							O:BPI_BUS23	1
							O:URTS2	3
							O:UDI_TDO_1	4
CAM_RST2	AP8	DIO	Reset control to 2nd sensor	DVDD18	PD	I	B:GPIO64	0
							O:DMIC1_CLK	1
							I0:UDI_TDI_2	4
CAM_RST3	AN8	DIO	Reset control to 3rd sensor	DVDD18	PD	I	B:GPIO65	0
							I0:DMIC1_DA T	1
							O:UDI_TDO_2	4
CAM_CLK0	AT7	DIO	Master clock to 0th sensor	DVDD18	PD	I	B:GPIO66	0
							O:CMMCLK0	1
							I0:UDI_TDI_3	4
							O:DBG_MON _B7	7
CAM_CLK1	AE1	DIO	Master clock to 1st sensor	DVDD18	PD	I	B:GPIO67	0
							O:CMMCLK1	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:UDI_TDO_3	4
							O:DBG_MON_B8	7
CAM_CLK2	AU7	DIO	Master clock to 2nd sensor	DVDD18	PD	I	B:GPIO68	0
							O:CMMCLK2	1
							IO:UDI_TDI_4	4
							O:DBG_MON_B9	7
CAM_CLK3	AE2	DIO	Master clock to 3rd sensor	DVDD18	PD	I	B:GPIO69	0
							O:CMMCLK3	1
							O:UDI_TDO_4	4
							O:DBG_MON_B10	7
I2C								
SCL0	U32	DIO	I2C0 clock	DVDD18	PD	I	B:GPIO121	0
							B1:SCL0	1
SDA0	V32	DIO	I2C0 data	DVDD18	PD	I	B:GPIO122	0
							B1:SDA0	1
SCL1	V34	DIO	I2C1 clock	DVDD18	PD	I	B:GPIO123	0
							B1:SCL1	1
SDA1	V33	DIO	I2C1 data	DVDD18	PD	I	B:GPIO124	0
							B1:SDA1	1
CAM_SCL2	AN10	DIO	I2C2 clock	DVDD18	PD	I	B:GPIO125	0
							B1:SCL2	1
							B1:SCP_SCL2	2
							IO:UDI_TDI_6	4
CAM_SDA2	AM10	DIO	I2C2 data	DVDD18	PD	I	B:GPIO126	0
							B1:SDA2	1
							B1:SCP_SDA2	2

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:UDI_TDO_6	4
SCL3	AN19	DIO	I2C3 clock	DVDD18	PD	I	B:GPIO127	0
							B1:SCL3	1
							O:mbistreade n_trigger	6
SDA3	AM19	DIO	I2C3 data	DVDD18	PD	I	B:GPIO128	0
							B1:SDA3	1
							O:mbistwrite en_trigger	6
CAM_SCL4	AT10	DIO	I2C4 clock	DVDD18	PD	I	B:GPIO129	0
							B1:SCL4	1
							B1:SCP_SCL4	2
							IO:UDI_TDI_7	4
CAM_SDA4	AP9	DIO	I2C4 data	DVDD18	PD	I	B:GPIO130	0
							B1:SDA4	1
							B1:SCP_SDA4	2
							O:UDI_TDO_7	4
CHG_SCL5	AM18	DIO	I2C5 clock	DVDD18	PD	I	B:GPIO131	0
							B1:SCL5	1
							B0:TP_GPIO0 _AO	4
CHG_SDA5	AN18	DIO	I2C5 data	DVDD18	PD	I	B:GPIO132	0
							B1:SDA5	1
							B0:TP_GPIO1 _AO	4
SCL6	AR18	DIO	I2C6 clock	DVDD18	PD	I	B:GPIO133	0
							B1:SCL6	1
SDA6	AT18	DIO	I2C6 data	DVDD18	PD	I	B:GPIO134	0
							B1:SDA6	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
CAM_SCL7	AN9	DIO	I2C7 clock	DVDD18	PD	I	B:GPIO135	0
							B1:SCL7	1
CAM_SDA7	AM9	DIO	I2C7 data	DVDD18	PD	I	B:GPIO136	0
							B1:SDA7	1
CAM_SCL8	AT8	DIO	I2C8 clock	DVDD18	PD	I	B:GPIO137	0
							B1:SCL8	1
CAM_SDA8	AU8	DIO	I2C8 data	DVDD18	PD	I	B:GPIO138	0
							B1:SDA8	1
CAM_SCL9	AT9	DIO	I2C9 clock	DVDD18	PD	I	B:GPIO139	0
							B1:SCL9	1
							O:UTXD2	2
CAM_SDA9	AR9	DIO	I2C9 data	DVDD18	PD	I	B:GPIO140	0
							B1:SDA9	1
							I1:URXD2	2
SCL10	AP11	DIO	I2C10 clock	DVDD18	PD	I	B:GPIO141	0
							B1:SCL10	1
							I1:UCTS2	2
							O:GPS_L1_ELNA_EN	3
SDA10	AR11	DIO	I2C10 data	DVDD18	PD	I	B:GPIO142	0
							B1:SDA10	1
							O:URTS2	2
							O:GPS_L5_ELNA_EN	3
SCL11	AU11	DIO	I2C11 clock	DVDD18	PD	I	B:GPIO143	0
							B1:SCL11	1
							O:UTXD4	2
							O:PWM_2	3

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:DISP_ERRO R0	4
							O:TP_URTS2_ VCORE	6
SDA11	AT11	DIO	I2C11 data	DVDD18	PD	I	B:GPIO144	0
							B1:SDA11	1
							I1:URXD4	2
							O:PWM_3	3
							O:DISP_ERRO R1	4
							I1:TP_UCTS2_ VCORE	6
SCL12	AR10	DIO	I2C12 clock	DVDD18	PD	I	B:GPIO145	0
							B1:SCL12	1
							O:UTXD5	2
							O:MD_UTXD1	3
							O:SSPM_UTX D_AO_VCORE	5
							O:TP_UTXD2_ VCORE	6
							O:CONN_BT_ TXD	7
SDA12	AU10	DIO	I2C12 data	DVDD18	PD	I	B:GPIO146	0
							B1:SDA12	1
							I1:URXD5	2
							I1:MD_URXD 1	3
							I1:SSPM_URX D_AO_VCORE	5
							I1:TP_URXD2_ VCORE	6
Connectivity								
ANT_SELO	V3	DIO	Antenna select signal	DVDD18	PD	I	B:GPIO94	0

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:ANT_SELO	1
							O:PCIE_PERST_N	2
							B0:TP_GPIO14_AO	4
ANT_SEL1	U4	DIO	Antenna select signal	DVDD18	PD	I	B:GPIO95	0
							O:ANT_SEL1	1
							B1:PCIE_WAKEN	2
							B0:TP_GPIO15_AO	4
ANT_SEL2	U1	DIO	Antenna select signal	DVDD18	PD	I	O:PWM_3	5
							B:GPIO96	0
							O:ANT_SEL2	1
							B1:PCIE_CLKREQN	2
ANT_SEL3	T5	DIO	Antenna select signal	DVDD18	PD	I	B0:TP_GPIO0_AO	4
							I0:EXT_FRAME_SYNC	6
							B:GPIO97	0
							O:ANT_SEL3	1
							O:TP_UTXD2_VLP	2
							I0:I2SIN6_DI	3
ANT_SEL4	T4	DIO	Antenna select signal	DVDD18	PD	I	B0:TP_GPIO1_AO	4
							O:PWM_2	5
							O:MD_UTXD1	6
							B:GPIO98	0
							O:ANT_SEL4	1
							I1:TP_URXD2_VLP	2

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							IO:I2SIN6_DI1	6
							B0:TP_GPIO2_AO	4
							I1:MD_URXD1	6
ANT_SEL5	T2	DIO	Antenna select signal	DVDD18	PD	I	B:GPIO99	0
							O:ANT_SEL5	1
							O:PTA_TXD	2
							IO:I2SIN6_DI2	3
							O:CONN_TCX_OENA_REQ	4
							O:AGPS_SYNC	6
ANT_SEL6	T3	DIO	Antenna select signal	DVDD18	PD	I	B:GPIO100	0
							O:ANT_SEL6	1
							I1:PTA_RXD	2
							IO:I2SIN6_DI3	3
							O:PWM_1	5
CONN_HRST_B_2	V4	DIO	CONNSYS reset2	DVDD18	PD	I	B:GPIO101	0
							O:ANT_SEL7	1
							O:CONN_HRST_B_2	2
							O:I2SIN6_MCK	3
CONN_HRST_B	V5	DIO	CONNSYS reset	DVDD18	PD	I	B:GPIO102	0
							O:CONN_HRST_B	1
							O:CONN_RST	2
							B0:TP_GPIO3_AO	4
CONN_TOP_CLK	R1	DIO	CONNSYS top 2-wire interface	DVDD18	PD	I	B:GPIO103	0
							O:CONN_TOP_CLK	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:UTXD3	3
							O:CONN_UAR THUB_UART_ TX	4
							B:GPIO104	0
CONN_TOP_D ATA	R2	DIO	CONNSYS top 2-wire interface	DVDD18	PD	I	B0:CONN_TO P_DATA	1
							I1:URXD3	3
							I1:CONN_UAR THUB_UART_ RX	4
							B:GPIO105	0
CONN_BT_CLK	R5	DIO	CONNSYS BT interface	DVDD18	PD	I	B0:CONN_BT _CLK	1
							IO:CONN_FAU LTB	2
							B0:TP_GPIO4 _AO	4
							B:GPIO106	0
CONN_BT_DAT A	R4	DIO	CONNSYS BT interface	DVDD18	PD	I	B0:CONN_BT _DATA	1
							O:CONN_PMI C_EN	2
							B:GPIO107	0
CONN_WB_PT A	R3	DIO	CONNSYS Wi-Fi/Bluetooth interface	DVDD18	PD	I	B0:CONN_WB _PTA	1
							O:I2SIN2_BCK	2
							O:TP_UTXD1_ VCORE	6
							B:GPIO108	0
CONN_WF_CT RLO	U6	DIO	CONNSYS Wi-Fi control signal	DVDD18	PD	I	B0:CONN_WF _CTRL0	1
							O:I2SIN2_LRC K	2

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							I1:TP_URXD1_VCORE	6
CONN_WF_CTL1	T8	DIO	CONNSYS Wi-Fi control signal	DVDD18	PD	I	B:GPIO109	0
							B0:CONN_WF_CTRL1	1
							I0:I2SIN2_DI	2
							O:TP_URTS1_VCORE	6
CONN_WF_CTL2	T6	DIO	CONNSYS Wi-Fi control signal	DVDD18	PD	I	B:GPIO110	0
							B0:CONN_WF_CTRL2	1
							O:I2SOUT2_DO	2
							I1:TP_UCTS1_VCORE	6
CONN_WF_CTL3	T7	DIO	CONNSYS Wi-Fi control signal	DVDD18	PD	I	B:GPIO111	0
							B0:CONN_WF_CTRL3	1
							O:CONN_TOP_CLK_2	2
CONN_WF_CTL4	R6	DIO	CONNSYS Wi-Fi control signal	DVDD18	PD	I	B:GPIO112	0
							B0:CONN_WF_CTRL4	1
							B0:CONN_TO_P_DATA_2	2
GPS_L1_ELNA_EN	V6	DIO	GPS L1 ELNA enable	DVDD18	PD	I	B:GPIO92	0
							O:GPS_L1_ELNA_EN	1
							O:BT_RST	2
							O:DBG_MON_B11	7
GPS_L5_ELNA_EN	V7	DIO	GPS L5 ELNA enable	DVDD18	PD	I	B:GPIO93	0
							O:GPS_L5_ELNA_EN	1
							O:WF_RST	2

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:DBG_MON_B12	7
System companion processor (SCP)								
SCP_SPI0_CK	AC5	DIO	System companion processor clock	DVDD18	PD	I	B:GPIO48	0
							O:SPI0_CLK	1
							O:SCP_SPI0_CK	2
							I0:MD_INT0	3
							I1:SPM_JTAG_TCK_VLP	4
							I0:CONN_BGF_MCU_TCK	5
SCP_SPI0_CSB	AD6	DIO	System companion processor chip select	DVDD18	PD	I	B:GPIO49	0
							O:SPI0_CSB	1
							O:SCP_SPI0_CS	2
							I0:MD_INT3	3
							I1:SPM_JTAG_TMS_VLP	4
							I1:CONN_BGF_MCU_TMS	5
SCP_SPI0_MO	AC3	DIO	System companion processor dataout	DVDD18	PD	I	B:GPIO50	0
							B0:SPI0_MO	1
							B0:SCP_SPI0_MO	2
							I0:MD_INT4	3
							B1:SPM_JTAG_TDO_VLP	4
							O:CONN_BGF_MCU_TDO	5
SCP_SPI0_MI	AC4	DIO	System companion processor datain	DVDD18	PD	I	B:GPIO51	0
							B0:SPI0_MI	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							B0:SCP_SPI0_MI	2
							I1:SPM_JTAG_TDI_VLP	4
							I0:CONN_BGF_MCU_TDI	5
SCP_SPI1_CK	AB1	DIO	System companion processor clock	DVDD18	PD	I	B:GPIO44	0
							O:SPI1_CLK	1
							O:SCP_SPI1_CK	2
							O:DMIC1_CLK	3
							B0:TP_GPIO13_AO	4
							O:TP_UTXD2_VLP	5
							O:UTXD2	6
SCP_SPI1_CSB	AB2	DIO	System companion processor chip select	DVDD18	PD	I	O:DBG_MON_A24	7
							B:GPIO45	0
							O:SPI1_CSB	1
							O:SCP_SPI1_CS	2
							I0:DMIC1_DATA	3
							B0:TP_GPIO14_AO	4
							I1:TP_URXD2_VLP	5
SCP_SPI1_MO	AB3	DIO	System companion processor dataout	DVDD18	PD	I	I1:URXD2	6
							O:DBG_MON_A25	7
							B:GPIO46	0
							B0:SPI1_MO	1
							B0:SCP_SPI1_MO	2

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:DMIC_CLK	3
							O:SCP_DMIC_CLK	4
							O:TP_URTS2_VLP	5
							I1:UCTS2	6
							O:DBG_MON_A26	7
SCP_SPI1_MI	AB4	DIO	System companion processor datain	DVDD18	PD	I	B:GPIO47	0
							B0:SPI1_MI	1
							B0:SCP_SPI1_MI	2
							I0:DMIC_DAT	3
							I0:SCP_DMIC_DAT	4
							I1:TP_UCTS2_VLP	5
							O:URTS2	6
							O:DBG_MON_A27	7
SCP_SPI2_CK	AA1	DIO	System companion processor clock	DVDD18	PD	I	B:GPIO40	0
							O:SPI2_CLK	1
							O:SCP_SPI2_CK	2
							B0:TP_GPIO9_AO	4
							I1:UCTS3	5
							O:TP_UTXD1_VLP	6
							O:DBG_MON_A20	7
SCP_SPI2_CSB	AA4	DIO	System companion processor chip select	DVDD18	PD	I	B:GPIO41	0
							O:SPI2_CSB	1

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:SCP_SPI2_C S	2
							B0:TP_GPIO1 0_AO	4
							O:URTS3	5
							I1:TP_URXD1 _VLP	6
							O:DBG_MON _A21	7
SCP_SPI2_MO	AA3	DIO	System companion processor dataout	DVDD18	PD	I	B:GPIO42	0
							B0:SPI2_MO	1
							B0:SCP_SPI2_ MO	2
							B0:TP_GPIO1 1_AO	4
							O:UTXD3	5
							O:TP_URTS1_ VLP	6
							O:DBG_MON _A22	7
SCP_SPI2_MI	AA5	DIO	System companion processor datain	DVDD18	PD	I	B:GPIO43	0
							B0:SPI2_MI	1
							B0:SCP_SPI2_ MI	2
							B0:TP_GPIO1 2_AO	4
							I1:URXD3	5
							I1:TP_UCTS1_ VLP	6
							O:DBG_MON _A23	7
SCP_SPI3_CLK	AD7	DIO	System companion processor clock	DVDD18	PD	I	B:GPIO36	0
							O:SPI3_CLK	1
							O:SCP_SPI3_C K	2

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							I0:MD_INT0	3
							B0:TP_GPIO5_AO	4
							O:UTXD4	5
							O:MD_UTXD0	6
							O:DBG_MON_A16	7
SCP_SPI3_CSB	AD8	DIO	System companion processor chip select	DVDD18	PD	I	B:GPIO37	0
							O:SPI3_CSB	1
							O:SCP_SPI3_CS	2
							I0:MD_INT3	3
							B0:TP_GPIO6_AO	4
							I1:URXD4	5
							I1:MD_URXD0	6
							O:DBG_MON_A17	7
SCP_SPI3_MO	AC8	DIO	System companion processor dataout	DVDD18	PD	I	B:GPIO38	0
							B0:SPI3_MO	1
							B0:SCP_SPI3_MO	2
							I0:MD_INT4	3
							B0:TP_GPIO7_AO	4
							O:UTXD5	5
SCP_SPI3_MI	AC7	DIO	System companion processor datain	DVDD18	PD	I	O:DBG_MON_A18	7
							B:GPIO39	0
							B0:SPI3_MI	1
							B0:SCP_SPI3_MI	2

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							B0:TP_GPIO8_AO	4
							I1:URXD5	5
							O:DBG_MON_A19	7
SCP_I2C_SCL0	Y1	DIO	System companion processor I2C0 CLK	DVDD18	PD	I	B:GPIO80	0
							B1:SCP_SCL0	1
							I1:UCTS3	2
							O:CMMCLK4	3
							B0:TP_GPIO2_AO	4
							O:PWM_VLP	5
SCP_I2C_SDA0	W1	DIO	System companion processor I2C0 DATA	DVDD18	PD	I	O:TP_UTXD2_VLP	6
							B:GPIO81	0
							B1:SCP_SDA0	1
							O:URTS3	2
							O:CMMCLK5	3
							B0:TP_GPIO3_AO	4
SCP_I2C_SCL1	W7	DIO	System companion processor I2C1 CLK	DVDD18	PD	I	I0:SRCLKENAI_1	5
							I1:TP_URXD2_VLP	6
							B:GPIO82	0
							B1:SCP_SCL1	1
							O:UTXD3	2
							O:CMMCLK4	3
							B0:TP_GPIO4_AO	4
							I0:MD_INT3	5

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:TP_URTS2_VLP	6
SCP_I2C_SDA1	W8	DIO	System companion processor I2C1 DATA	DVDD18	PD	I	B:GPIO83	0
							B1:SCP_SDA1	1
							I1:URXD3	2
							O:CMMCLK5	3
							B0:TP_GPIO5_AO	4
							I0:MD_INT4	5
SCP_I2C_SCL2	W5	DIO	System companion processor I2C2 CLK	DVDD18	PD	I	I1:TP_UCTS2_VLP	6
							B:GPIO84	0
							B1:SCP_SCL2	1
							O:MD_MCIF_UTXD0	2
							O:MD_UTXD0	3
							B0:TP_GPIO6_AO	4
SCP_I2C_SDA2	W2	DIO	System companion processor I2C2 DATA	DVDD18	PD	I	O:SSPM_UTXD_AO_VLP	5
							O:TP_UTXD1_VLP	6
							O:CONN_BGF_UART0_TXD	7
							B:GPIO85	0
							B1:SCP_SDA2	1
							I1:MD_MCIF_URXD0	2
							I1:MD_URXD0	3
							B0:TP_GPIO7_AO	4
							I1:SSPM_URXD_AO_VLP	5

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							I1:TP_URXD1_VLP	6
							I1:CONN_BGF_UART0_RXD	7
SCP_I2C_SCL3	W3	DIO	System companion processor I2C3 CLK	DVDD18	PD	I	B:GPIO86	0
							B1:SCP_SCL3	1
							B0:I2SIN6_BCK	3
							B0:TP_GPIO8_AO	4
							IO:EXTIF0_ACT	5
							O:TP_URTS1_VLP	6
SCP_I2C_SDA3	V2	DIO	System companion processor I2C3 DATA	DVDD18	PD	I	B:GPIO87	0
							B1:SCP_SDA3	1
							O:I2SOUT6_DO	3
							B0:TP_GPIO9_AO	4
							IO:EXTIF0_PRI	5
							I1:TP_UCTS1_VLP	6
SCP_I2C_SCL4	Y3	DIO	System companion processor I2C4 CLK	DVDD18	PD	I	B:GPIO88	0
							B1:SCP_SCL4	1
							O:I2SOUT6_DO1	3
							B0:TP_GPIO10_AO	4
							O:EXTIF0_GNT_B	5
SCP_I2C_SDA4	Y2	DIO	System companion processor I2C4 DATA	DVDD18	PD	I	B:GPIO89	0
							B1:SCP_SDA4	1
							O:I2SOUT6_DO2	3

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							B0:TP_GPIO11_AO	4
							I0:MD_INT0	5
SCP_I2C_SCL5	AB7	DIO	System companion processor I2C5 CLK	DVDD18	PD	I	B:GPIO90	0
							B1:SCP_SCL5	1
							B0:I2SIN6_LRCK	3
							B0:TP_GPIO12_AO	4
							O:MD_GPS_L1_BLANK	6
SCP_I2C_SDA5	AB8	DIO	System companion processor I2C5 DATA	DVDD18	PD	I	B:GPIO91	0
							B1:SCP_SDA5	1
							O:I2SOUT6_D03	3
							B0:TP_GPIO13_AO	4
							O:MD_GPS_L5_BLANK	6
SCP_VREQ_VAO	AT17	DIO	System companion processor voltage request	DVDD18	PD	I	B:GPIO185	0
							O:SCP_VREQ_VAO	1
General purpose								
PERI_EN0	AN13	DIO	PERIPHERAL_EN0	DVDD18	PD	OL	B:GPIO113	0
							O:PWM_0	1
							O:MD1_SIM2_SCLK	2
PERI_EN1	AM13	DIO	PERIPHERAL_EN1	DVDD18	PD	OL	B:GPIO114	0
							O:PWM_1	1
							O:MD1_SIM2_SRST	2
							O:TP_UTXD2_VCORE	5

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:UTXD2	6
							O:DBG_MON_B13	7
PERI_EN2	AP13	DIO	PERIPHERAL_EN2	DVDD18	PD	OL	B:GPIO115	0
							O:PWM_2	1
							B1:MD1_SIM2_SIO	2
							I1:TP_URXD2_VCORE	5
							I1:URXD2	6
							O:DBG_MON_B14	7
PERI_EN3	AP14	DIO	PERIPHERAL_EN3	DVDD18	PD	OL	B:GPIO116	0
							O:PWM_3	1
							O:MD1_SIM1_SCLK	2
PERI_EN4	AR13	DIO	PERIPHERAL_EN4	DVDD18	PD	OL	B:GPIO117	0
							I0:DSI_TE1	1
							O:MD1_SIM1_SRST	2
							I0:UDI_TDI_5	4
							O:MD_GPS_L1_BLANK	6
							O:DBG_MON_A30	7
PERI_EN5	AT14	DIO	PERIPHERAL_EN5	DVDD18	PD	OL	B:GPIO118	0
							O:LCM_RST1	1
							B1:MD1_SIM1_SIO	2
							O:UDI_TDO_5	4
							O:MD_GPS_L5_BLANK	6

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
							O:DBG_MON_A31	7
JTAG								
JTRSTN	T32	DIO	JTAG interface	DVDD18	PU	I	B:GPIO169	0
							I1:JTRSTN_SE L1_VCORE	1
							I0:SPM_JTAG_TRSTN_VCORE	6
JTDO	T34	DIO	JTAG interface	DVDD18	PD	I	B:GPIO170	0
							O:JTDO_SEL1_VCORE	1
							B1:SPM_JTAG_TDO_VCORE	6
JTCK	T33	DIO	JTAG interface	DVDD18	PD	I	B:GPIO171	0
							I0:JTCK_SEL1_VCORE	1
							I1:SPM_JTAG_TCK_VCORE	6
JTMS	U33	DIO	JTAG interface	DVDD18	PD	I	B:GPIO172	0
							B0:JTMS_SEL1_VCORE	1
							I1:SPM_JTAG_TMS_VCORE	6
JTDI	U34	DIO	JTAG interface	DVDD18	PD	I	B:GPIO173	0
							I0:JTDI_SEL1_VCORE	1
							I1:SPM_JTAG_TDI_VCORE	6
Digital-RF (DRF)								
DRF_DL_CP	AK30	AIO	DRF interface	AVDD12_DRF				
DRF_DL_CN	AK31	AIO	DRF interface	AVDD12_DRF				
DRF_DL_LNO_DP	AL34	AIO	DRF interface	AVDD12_DRF				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
DRF_DL_LN0_DN	AK34	AIO	DRF interface	AVDD12_DRF				
DRF_DL_LN1_DP	AJ32	AIO	DRF interface	AVDD12_DRF				
DRF_DL_LN1_DN	AJ33	AIO	DRF interface	AVDD12_DRF				
DRF_DL_LN2_DP	AH30	AIO	DRF interface	AVDD12_DRF				
DRF_DL_LN2_DN	AH31	AIO	DRF interface	AVDD12_DRF				
DRF_DL_LN3_DP	AH34	AIO	DRF interface	AVDD12_DRF				
DRF_DL_LN3_DN	AG34	AIO	DRF interface	AVDD12_DRF				
DRF_UL_CP	AM33	AIO	DRF interface	AVDD12_DRF				
DRF_UL_CN	AM32	AIO	DRF interface	AVDD12_DRF				
DRF_UL_LN1_DP	AN31	AIO	DRF interface	AVDD12_DRF				
DRF_UL_LN1_DN	AN30	AIO	DRF interface	AVDD12_DRF				
DRF_UL_LN0_DP	AP33	AIO	DRF interface	AVDD12_DRF				
DRF_UL_LN0_DN	AR33	AIO	DRF interface	AVDD12_DRF				
MIPI								
DSI0_D3N_T3A	AB32	AIO	DSI interface	AVDD12_DSI				
DSI0_D3P_T2C	AB33	AIO	DSI interface	AVDD12_DSI				
DSI0_D1N_T2B	AB34	AIO	DSI interface	AVDD12_DSI				
DSI0_D1P_T2A	AC34	AIO	DSI interface	AVDD12_DSI				
DSI0_CK0N_T1C	AB30	AIO	DSI interface	AVDD12_DSI				
DSI0_CK0P_T1B	AB31	AIO	DSI interface	AVDD12_DSI				
DSI0_D0N_T1A	AC30	AIO	DSI interface	AVDD12_DSI				
DSI0_D0P_T0C	AC31	AIO	DSI interface	AVDD12_DSI				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
DSI0_D2N_T0B	AC33	AIO	DSI interface	AVDD12_DSI				
DSI0_D2P_T0A	AC32	AIO	DSI interface	AVDD12_DSI				
DSI1_D3N_T3A	AD33	AIO	DSI interface	AVDD12_DSI				
DSI1_D3P_T2C	AD32	AIO	DSI interface	AVDD12_DSI				
DSI1_D1N_T2B	AE34	AIO	DSI interface	AVDD12_DSI				
DSI1_D1P_T2A	AE33	AIO	DSI interface	AVDD12_DSI				
DSI1_CK0N_T1 C	AE30	AIO	DSI interface	AVDD12_DSI				
DSI1_CK0P_T1 B	AE31	AIO	DSI interface	AVDD12_DSI				
DSI1_D0N_T1A	AF30	AIO	DSI interface	AVDD12_DSI				
DSI1_D0P_T0C	AF31	AIO	DSI interface	AVDD12_DSI				
DSI1_D2N_T0B	AF33	AIO	DSI interface	AVDD12_DSI				
DSI1_D2P_T0A	AF32	AIO	DSI interface	AVDD12_DSI				
CSI0B_L1N	AP6	AIO	CSI interface	AVDD12_CSI				
CSI0B_L1P_T0 C	AP5	AIO	CSI interface	AVDD12_CSI				
CSI0B_L0N_T0 B	AR5	AIO	CSI interface	AVDD12_CSI				
CSI0B_L0P_T0 A	AR4	AIO	CSI interface	AVDD12_CSI				
CSI0A_L2N_T1 C	AU5	AIO	CSI interface	AVDD12_CSI				
CSI0A_L2P_T1 B	AU4	AIO	CSI interface	AVDD12_CSI				
CSI0A_L1N_T1 A	AU3	AIO	CSI interface	AVDD12_CSI				
CSI0A_L1P_T0 C	AT3	AIO	CSI interface	AVDD12_CSI				
CSI0A_L0N_T0 B	AT2	AIO	CSI interface	AVDD12_CSI				
CSI0A_L0P_T0 A	AR3	AIO	CSI interface	AVDD12_CSI				
CSI1B_L1N	AR2	AIO	CSI interface	AVDD12_CSI				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
CSI1B_L1P_T0 C	AP3	AIO	CSI interface	AVDD12_CSI				
CSI1B_L0N_T0 B	AP2	AIO	CSI interface	AVDD12_CSI				
CSI1B_L0P_T0 A	AP1	AIO	CSI interface	AVDD12_CSI				
CSI1A_L2N_T1 C	AN5	AIO	CSI interface	AVDD12_CSI				
CSI1A_L2P_T1 B	AN4	AIO	CSI interface	AVDD12_CSI				
CSI1A_L1N_T1 A	AM5	AIO	CSI interface	AVDD12_CSI				
CSI1A_L1P_T0 C	AM4	AIO	CSI interface	AVDD12_CSI				
CSI1A_L0N_T0 B	AM3	AIO	CSI interface	AVDD12_CSI				
CSI1A_L0P_T0 A	AM2	AIO	CSI interface	AVDD12_CSI				
CSI2B_L1N	AL2	AIO	CSI interface	AVDD12_CSI				
CSI2B_L1P_T0 C	AL1	AIO	CSI interface	AVDD12_CSI				
CSI2B_L0N_T0 B	AK2	AIO	CSI interface	AVDD12_CSI				
CSI2B_L0P_T0 A	AK1	AIO	CSI interface	AVDD12_CSI				
CSI2A_L2N_T1 C	AL4	AIO	CSI interface	AVDD12_CSI				
CSI2A_L2P_T1 B	AL3	AIO	CSI interface	AVDD12_CSI				
CSI2A_L1N_T1 A	AK5	AIO	CSI interface	AVDD12_CSI				
CSI2A_L1P_T0 C	AK4	AIO	CSI interface	AVDD12_CSI				
CSI2A_L0N_T0 B	AJ4	AIO	CSI interface	AVDD12_CSI				
CSI2A_L0P_T0 A	AJ5	AIO	CSI interface	AVDD12_CSI				
CSI3B_L1N	AJ2	AIO	CSI interface	AVDD12_CSI				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
CSI3B_L1P_T0 C	AJ3	AIO	CSI interface	AVDD12_CSI				
CSI3B_L0N_T0 B	AH2	AIO	CSI interface	AVDD12_CSI				
CSI3B_L0P_T0 A	AH3	AIO	CSI interface	AVDD12_CSI				
CSI3A_L2N_T1 C	AG1	AIO	CSI interface	AVDD12_CSI				
CSI3A_L2P_T1 B	AG2	AIO	CSI interface	AVDD12_CSI				
CSI3A_L1N_T1 A	AG3	AIO	CSI interface	AVDD12_CSI				
CSI3A_L1P_T0 C	AG5	AIO	CSI interface	AVDD12_CSI				
CSI3A_L0N_T0 B	AF4	AIO	CSI interface	AVDD12_CSI				
CSI3A_L0P_T0 A	AF3	AIO	CSI interface	AVDD12_CSI				
WBG (Wi-Fi/Bluetooth/GPS)								
XIN_WBG	L10	AIO	WBG crystal clock input	AVDD15_WBG				
GL1_I	K5	AIO	GPS L1 I channel	AVDD15_WBG				
GL1_Q	K4	AIO	GPS L1 Q channel	AVDD15_WBG				
GL5_I	K3	AIO	GPS L5 I channel	AVDD15_WBG				
GL5_Q	L3	AIO	GPS L5 Q channel	AVDD15_WBG				
BT_IN	J1	AIO	BT I channel negative-end	AVDD15_WBG				
BT_IP	H1	AIO	BT I channel positive-end	AVDD15_WBG				
BT_QN	K2	AIO	BT Q channel negative-end	AVDD15_WBG				
BT_QP	J2	AIO	BT Q channel positive-end	AVDD15_WBG				
WF0_IN	G5	AIO	Wi-Fi I channel 0 negative-end	AVDD15_WBG				
WF0_IP	G6	AIO	Wi-Fi I channel 0 positive-end	AVDD15_WBG				
WF0_QN	H6	AIO	Wi-Fi Q channel 0 negative-end	AVDD15_WBG				
WF0_QP	H5	AIO	Wi-Fi Q channel 0 positive-end	AVDD15_WBG				
WF1_IN	F3	AIO	Wi-Fi I channel 1 negative-end	AVDD15_WBG				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
WF1_IP	F4	AIO	Wi-Fi I channel 1 positive-end	AVDD15_WBG				
WF1_QN	G2	AIO	Wi-Fi Q channel 1 negative-end	AVDD15_WBG				
WF1_QP	F2	AIO	Wi-Fi Q channel 1 positive-end	AVDD15_WBG				
USB								
USB_DM	J31	AIO	USB D- differential data line	AVDD33_USB				
USB_DP	J30	AIO	USB D+ differential data line	AVDD33_USB				
SSUSB								
SSUSB_TXP_P1	L30	AIO	SSUSB TXDATA	AVDD12_SSUSB				
SSUSB_TXN_P1	L31	AIO	SSUSB TXDATA	AVDD12_SSUSB				
SSUSB_RXP_P1	K29	AIO	SSUSB RXDATA	AVDD12_SSUSB				
SSUSB_RXN_P1	K28	AIO	SSUSB RXDATA	AVDD12_SSUSB				
PCIe								
CKM_XTAL_CK	N11	AIO	26M reference clock input for PCIe	AVDD12_PCIE				
PCIE_CKP	M1	AIO	PCIe interface	AVDD12_PCIE				
PCIE_CKN	M2	AIO	PCIe interface	AVDD12_PCIE				
PCIE_LN0_TXP	N5	AIO	PCIe interface	AVDD12_PCIE				
PCIE_LN0_TXN	N4	AIO	PCIe interface	AVDD12_PCIE				
PCIE_LN0_RXP	M6	AIO	PCIe interface	AVDD12_PCIE				
PCIE_LN0_RXN	M7	AIO	PCIe interface	AVDD12_PCIE				
UFS								
UFS_PLL_CKREF	E34	AIO	26 MHz clock input for UFS	AVDD12_UFS				
UFS_REFCK_OUT	H28	AIO	UFS interface	AVDD12_UFS				
UFS_RST_N	F33	AIO	UFS interface	AVDD12_UFS				
UFS_RX0N	C33	AIO	UFS interface	AVDD12_UFS				
UFS_RX0P	C34	AIO	UFS interface	AVDD12_UFS				
UFS_RX1N	D32	AIO	UFS interface	AVDD12_UFS				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
UFS_RX1P	E32	AIO	UFS interface	AVDD12_UFS				
UFS_TX0N	G33	AIO	UFS interface	AVDD12_UFS				
UFS_TX0P	G34	AIO	UFS interface	AVDD12_UFS				
UFS_TX1N	G30	AIO	UFS interface	AVDD12_UFS				
UFS_TX1P	G31	AIO	UFS interface	AVDD12_UFS				
MISC								
TN_APPLLGP	T19	AIO	Reserved					
TP_APPLLGP	R19	AIO	Reserved					
PWRGD_TRIM	J23	AIO	Reserved					
EMI								
EMIO_CA0	A11	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CA1	A9	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CA2	B8	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CA3	D9	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CA4	B12	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CA5	C9	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CA6	D10	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CK_C	D7	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CK_T	C7	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CS1	E8	AIO	DRAM interface	AVDDQ_EMI				
EMIO_CS0	E9	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DMIO	B14	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DMI1	C4	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ0	A15	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ1	B16	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ2	A17	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ3	D13	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ4	C13	AIO	DRAM interface	AVDDQ_EMI				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
EMIO_DQ5	A13	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ6	D12	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ7	D11	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ8	A3	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ9	B4	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ10	A5	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ11	D5	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ12	B6	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ13	E6	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ14	A7	AIO	DRAM interface	AVDDQ_EMI				
EMIO_DQ15	D3	AIO	DRAM interface	AVDDQ_EMI				
EMIO_RDQS1_C	C1	AIO	DRAM interface	AVDDQ_EMI				
EMIO_RDQS1_T	D1	AIO	DRAM interface	AVDDQ_EMI				
EMIO_RDQS0_C	F15	AIO	DRAM interface	AVDDQ_EMI				
EMIO_RDQS0_T	E15	AIO	DRAM interface	AVDDQ_EMI				
EMIO_WCK0_C	C14	AIO	DRAM interface	AVDDQ_EMI				
EMIO_WCK0_T	D14	AIO	DRAM interface	AVDDQ_EMI				
EMIO_WCK1_C	D2	AIO	DRAM interface	AVDDQ_EMI				
EMIO_WCK1_T	C2	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CA0	B24	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CA1	B26	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CA2	E22	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CA3	A27	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CA4	A23	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CA5	C21	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CA6	E21	AIO	DRAM interface	AVDDQ_EMI				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
NC_EMI1_CA7		AIO	DRAM interface	AVDDQ_EMI				
EMI1_CK_C	C24	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CK_T	D24	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CS1	F24	AIO	DRAM interface	AVDDQ_EMI				
EMI1_CS0	F23	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DMI0	C27	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DMI1	A21	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ0	B32	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ1	A31	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ2	B30	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ3	D27	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ4	A29	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ5	D26	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ6	B28	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ7	C28	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ8	B20	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ9	A19	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ10	B18	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ11	C19	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ12	D18	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ13	B22	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ14	E19	AIO	DRAM interface	AVDDQ_EMI				
EMI1_DQ15	D20	AIO	DRAM interface	AVDDQ_EMI				
EMI1_RDQS0_C	C30	AIO	DRAM interface	AVDDQ_EMI				
EMI1_RDQS0_T	D30	AIO	DRAM interface	AVDDQ_EMI				
EMI1_RDQS1_C	F16	AIO	DRAM interface	AVDDQ_EMI				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
EMI1_RDQS1_T	E16	AIO	DRAM interface	AVDDQ_EMI				
EMI1_WCK1_T	C17	AIO	DRAM interface	AVDDQ_EMI				
EMI1_WCK1_C	D17	AIO	DRAM interface	AVDDQ_EMI				
EMI1_WCK0_C	D29	AIO	DRAM interface	AVDDQ_EMI				
EMI1_WCK0_T	C29	AIO	DRAM interface	AVDDQ_EMI				
EMI_EXTR	E1	AIO	DRAM interface	AVDDQ_EMI				
EMI_RESET_N	C31	AIO	DRAM interface	AVDDQ_EMI				
Analog power								
AVDD12_USB	L34	P	Analog power for USB	AVDD12_USB				
AVDD15_USB	K33	P	Analog power for USB	AVDD15_USB				
AVDD33_USB	L33	P	Analog power for USB	AVDD33_USB				
AVDD12_SSUS_B_P1	M34	P	Analog power for SSUSB	AVDD12_SSUSB_P1				
AVDD15_SSUS_B_P1	N34	P	Analog power for SSUSB	AVDD15_SSUSB_P1				
AVDD12_CKSQ	AA29	P	Analog power for clock square	AVDD12_CKSQ				
AVDD15_CKSQ	AA27	P	Analog power for clock square	AVDD15_CKSQ				
AVDD08_DRV_DSI	AE29	P	Analog power for DSI	AVDD08_DRV_DSI				
AVDD12_DSI	AC28	P	Analog power for DSI	AVDD12_DSI				
AVDD15_DSI	AF28	P	Analog power for DSI	AVDD15_DSI				
AVDD12_DRF	AN34	P	Analog power for DRF	AVDD12_DRF				
AVDD15_DRF	AJ28	P	Analog power for DRF	AVDD15_DRF				
AVDD12_CSI	AU6	P	Analog power for CSI	AVDD12_CSI				
AVDD12_PCIE	P7	P	Analog power for PCIe	AVDD12_PCIE				
AVDD15_PCIE	p8	P	Analog power for PCIe	AVDD15_PCIE				
AVDD12_WBG	K7	P	Analog power for WBG (Wi-Fi/BT/GPS)	AVDD12_WBG				
AVDD15_WBG	F1	P	Analog power for WBG (Wi-Fi/BT/GPS)	AVDD15_WBG				
AVDD12_EMI	K12	P	DDRPHY power	AVDD12_EMI				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
AVDD15_EMI	J12	P	DDRPHY power	AVDD15_EMI				
AVDDQ_EMI0	K13	P	DDRPHY power	AVDDQ_EMI0				
	K15	P	DDRPHY power	AVDDQ_EMI0				
AVDD075_EMI0	K14	P	DDRPHY power	AVDD075_EMI0				
	K16	P	DDRPHY power	AVDD075_EMI0				
AVDDQ_EMI1	K19	P	DDRPHY power	AVDDQ_EMI1				
	K21	P	DDRPHY power	AVDDQ_EMI1				
AVDD075_EMI1	K18	P	DDRPHY power	AVDD075_EMI1				
	K20	P	DDRPHY power	AVDD075_EMI1				
VDD2H_EMI	L17	P	DRAM power	VDD2H_EMI				
AVDD12_CKBUFF_UFS	H27	P	Analog power for UFS	AVDD12_CKBUFF_UFS				
AVDD12_UFS	K24	P	Analog power for UFS	AVDD12_UFS				
AVDD15_UFS	J24	P	Analog power for UFS	AVDD15_UFS				
AVDD12_MCUPLL	T18	P	Analog power for MCU PLL	AVDD12_MCUPLL				
AVDD15_MCUPLL	R18	P	Analog power for MCU PLL	AVDD15_MCUPLL				
AVDD12_PLL	AA19	P	Analog power for AP PLL	AVDD12_PLL				
AVDD15_PLL	Y19	P	Analog power for AP PLL	AVDD15_PLL				
AVDD12_GPUPLL	AD14	P	Analog power for GPU PLL	AVDD12_GPUPLL				
AVDD15_GPUPLL	AE15	P	Analog power for GPU PLL	AVDD15_GPUPLL				
AVDD12_NPUPLL	P12	P	Analog power for NPU PLL	AVDD12_NPUPLL				
AVDD15_NPUPLL	R11	P	Analog power for NPU PLL	AVDD15_NPUPLL				
AVDD12_MDPLL	AC19	P	Analog power for MD PLL	AVDD12_MDPLL				
AVDD15_MDPLL	AD19	P	Analog power for MD PLL	AVDD15_MDPLL				
AVDD15_PROC	R20	P	Analog power for CPU	AVDD15_PROC				

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
AVDD15_PWRGD	H23	P	Analog power for PWRGD	AVDD15_PWRGD				
Digital power								
DVDD_CORE	L23	P	Digital power input for core					
	M9	P	Digital power input for core					
	M12	P	Digital power input for core					
	M17	P	Digital power input for core					
	N13	P	Digital power input for core					
	N15	P	Digital power input for core					
	N17	P	Digital power input for core					
	R13	P	Digital power input for core					
	R15	P	Digital power input for core					
	T15	P	Digital power input for core					
	T17	P	Digital power input for core					
	U17	P	Digital power input for core					
	V15	P	Digital power input for core					
	V17	P	Digital power input for core					
	W14	P	Digital power input for core					
	Y9	P	Digital power input for core					
	Y12	P	Digital power input for core					
	Y15	P	Digital power input for core					
	Y17	P	Digital power input for core					
	AE14	P	Digital power input for core					
	AL10	P	Digital power input for core					
DVDD_GPU ACK	AA14	P	Digital power input for GPU					
	AB16	P	Digital power input for GPU					
	AC14	P	Digital power input for GPU					
	AC18	P	Digital power input for GPU					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	AD16	P	Digital power input for GPU					
	AE16	P	Digital power input for GPU					
	AF16	P	Digital power input for GPU					
	AF18	P	Digital power input for GPU					
	AG14	P	Digital power input for GPU					
	AG18	P	Digital power input for GPU					
	AH16	P	Digital power input for GPU					
	AJ14	P	Digital power input for GPU					
	AJ15	P	Digital power input for GPU					
	AJ16	P	Digital power input for GPU					
	AJ18	P	Digital power input for GPU					
DVDD_MM	AB10	P	Digital power input for multimedia					
	AC10	P	Digital power input for multimedia					
	AC12	P	Digital power input for multimedia					
	AC13	P	Digital power input for multimedia					
	AD10	P	Digital power input for multimedia					
	AE12	P	Digital power input for multimedia					
	AF12	P	Digital power input for multimedia					
	AG12	P	Digital power input for multimedia					
	AH11	P	Digital power input for multimedia					
	AJ12	P	Digital power input for multimedia					
	AJ13	P	Digital power input for multimedia					
	AK10	P	Digital power input for multimedia					
DVDD_MODE M	AB19	P	Digital power input for modem					
	AB21	P	Digital power input for modem					
	AB23	P	Digital power input for modem					
	AB25	P	Digital power input for modem					
	AC21	P	Digital power input for modem					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	AD21	P	Digital power input for modem					
	AD23	P	Digital power input for modem					
	AD25	P	Digital power input for modem					
	AE19	P	Digital power input for modem					
	AF21	P	Digital power input for modem					
	AF23	P	Digital power input for modem					
	AF25	P	Digital power input for modem					
	AG19	P	Digital power input for modem					
	AG21	P	Digital power input for modem					
	AG25	P	Digital power input for modem					
	AH23	P	Digital power input for modem					
	AJ19	P	Digital power input for modem					
	AJ23	P	Digital power input for modem					
	AJ25	P	Digital power input for modem					
	AK26	P	Digital power input for modem					
DVDD_NPU	R10	P	Digital power input for NPU					
	T10	P	Digital power input for NPU					
	T11	P	Digital power input for NPU					
	U10	P	Digital power input for NPU					
	U12	P	Digital power input for NPU					
	U13	P	Digital power input for NPU					
	V10	P	Digital power input for NPU					
	V12	P	Digital power input for NPU					
	W10	P	Digital power input for NPU					
DVDD_PROC_B	M25	P	Digital power input for big core					
	N25	P	Digital power input for big core					
	P25	P	Digital power input for big core					
	R25	P	Digital power input for big core					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	T25	P	Digital power input for big core					
	T26	P	Digital power input for big core					
	T27	P	Digital power input for big core					
	T28	P	Digital power input for big core					
	T29	P	Digital power input for big core					
	T31	P	Digital power input for big core					
	U25	P	Digital power input for big core					
	U29	P	Digital power input for big core					
	U31	P	Digital power input for big core					
	V31	P	Digital power input for big core					
	W21	P	Digital power input for big core					
	W25	P	Digital power input for big core					
	W26	P	Digital power input for big core					
	W27	P	Digital power input for big core					
	W28	P	Digital power input for big core					
	W29	P	Digital power input for big core					
	W31	P	Digital power input for big core					
	Y21	P	Digital power input for big core					
	Y25	P	Digital power input for big core					
	Y29	P	Digital power input for big core					
	Y31	P	Digital power input for big core					
	AA21	P	Digital power input for big core					
	AA23	P	Digital power input for big core					
	AA25	P	Digital power input for big core					
DVDD_PROC_L	N19	P	Digital power input for little core					
	N21	P	Digital power input for little core					
	N23	P	Digital power input for little core					
	P19	P	Digital power input for little core					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	P21	P	Digital power input for little core					
	R23	P	Digital power input for little core					
	T21	P	Digital power input for little core					
	T23	P	Digital power input for little core					
	U19	P	Digital power input for little core					
	U21	P	Digital power input for little core					
	V19	P	Digital power input for little core					
	V21	P	Digital power input for little core					
	V23	P	Digital power input for little core					
	W19	P	Digital power input for little core					
DVDD_SRAM_CORE	M15	P	Digital power input for core SRAM					
	M19	P	Digital power input for core SRAM					
	P14	P	Digital power input for core SRAM					
	U14	P	Digital power input for core SRAM					
	V14	P	Digital power input for core SRAM					
	Y14	P	Digital power input for core SRAM					
	Y16	P	Digital power input for core SRAM					
	AK18	P	Digital power input for core SRAM					
DVDD_SRAM_CORE_1	M27	P	Digital power input for core SRAM					
DVDD_SRAM_GPUSTACK	AC16	P	Digital power input for GPU SRAM					
	AG16	P	Digital power input for GPU SRAM					
DVDD_SRAM_MM	AB12	P	Digital power input for multimedia SRAM					
	AD12	P	Digital power input for multimedia SRAM					
	AE11	P	Digital power input for multimedia SRAM					
	AH12	P	Digital power input for multimedia SRAM					
	AC23	P	Digital power input for modem SRAM					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
DVDD_SRAM_MODEM	AD20	P	Digital power input for modem SRAM					
	AG20	P	Digital power input for modem SRAM					
	AG23	P	Digital power input for modem SRAM					
DVDD_SRAM_NPU	R9	P	Digital power input for NPU SRAM					
DVDD_SRAM_PROC_B	R24	P	Digital power input for big core SRAM					
	U24	P	Digital power input for big core SRAM					
	Y23	P	Digital power input for big core SRAM					
DVDD_SRAM_PROC_L	R21	P	Digital power input for little core SRAM					
	W20	P	Digital power input for little core SRAM					
DVDD_SRAM_SCP	M21	P	Digital power input for SCP SRAM					
	M23	P	Digital power input for SCP SRAM					
DVDD18_VQPS	AA10	P	eFuse blowing power control					
DVDD12_ESIM	AU13	P	Digital power input for ESIM					
DVDD18_ESIM	AR14	P	Digital power input for ESIM					
DVDD15_MSD C1	AR17	P	Digital power input for MSDC1					
DVDD28_MSD C1	AP17	P	Digital power input for MSDC1					
DVDD15_SIM	AU17	P	Digital power input for SIM1/SIM2					
DVDD28_SIM1	AU14	P	Digital power input for SIM1					
DVDD28_SIM2	AT13	P	Digital power input for SIM2					
DVDD12_IOB M_MIPI	AT31	P	Digital power input for I/O					
DVDD18_IOB M_MIPI	AR29	P	Digital power input for I/O					
DVDD18_IOB M	AP20	P	Digital power input for I/O					
	AP21	P	Digital power input for I/O					
DVDD18_IOLM	R31	P	Digital power input for I/O					
DVDD18_IORB	AP10	P	Digital power input for I/O					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
DVDD12_IORT_HS	P2	P	Digital power input for I/O					
DVDD18_IORT_HS	U5	P	Digital power input for I/O					
DVDD18_IORT	W4	P	Digital power input for I/O					
	Y5	P	Digital power input for I/O					
Ground								
DVSS	B2	G	Ground					
	B3	G	Ground					
	B5	G	Ground					
	B7	G	Ground					
	B9	G	Ground					
	B11	G	Ground					
	B13	G	Ground					
	B15	G	Ground					
	B17	G	Ground					
	B19	G	Ground					
	B21	G	Ground					
	B23	G	Ground					
	B25	G	Ground					
	B27	G	Ground					
	B29	G	Ground					
	B31	G	Ground					
	B33	G	Ground					
	C3	G	Ground					
	C5	G	Ground					
	C6	G	Ground					
	C10	G	Ground					
	C16	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	C18	G	Ground					
	C20	G	Ground					
	C25	G	Ground					
	C26	G	Ground					
	C32	G	Ground					
	D4	G	Ground					
	D6	G	Ground					
	D16	G	Ground					
	D19	G	Ground					
	D21	G	Ground					
	D25	G	Ground					
	D31	G	Ground					
	D33	G	Ground					
	D34	G	Ground					
	E2	G	Ground					
	E3	G	Ground					
	E4	G	Ground					
	E5	G	Ground					
	E7	G	Ground					
	E10	G	Ground					
	E13	G	Ground					
	E14	G	Ground					
	E23	G	Ground					
	E24	G	Ground					
	E25	G	Ground					
	E27	G	Ground					
	E28	G	Ground					
	E29	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	E30	G	Ground					
	E31	G	Ground					
	E33	G	Ground					
	F5	G	Ground					
	F6	G	Ground					
	F7	G	Ground					
	F8	G	Ground					
	F9	G	Ground					
	F10	G	Ground					
	F14	G	Ground					
	F17	G	Ground					
	F20	G	Ground					
	F25	G	Ground					
	F26	G	Ground					
	F27	G	Ground					
	F28	G	Ground					
	F29	G	Ground					
	F30	G	Ground					
	F31	G	Ground					
	F32	G	Ground					
	F34	G	Ground					
	G3	G	Ground					
	G4	G	Ground					
	G7	G	Ground					
	G8	G	Ground					
	G9	G	Ground					
	G10	G	Ground					
	G11	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	G15	G	Ground					
	G16	G	Ground					
	G17	G	Ground					
	G18	G	Ground					
	G19	G	Ground					
	G20	G	Ground					
	G21	G	Ground					
	G22	G	Ground					
	G23	G	Ground					
	G24	G	Ground					
	G25	G	Ground					
	G26	G	Ground					
	G27	G	Ground					
	G28	G	Ground					
	G29	G	Ground					
	G32	G	Ground					
	H2	G	Ground					
	H3	G	Ground					
	H4	G	Ground					
	H7	G	Ground					
	H8	G	Ground					
	H9	G	Ground					
	H10	G	Ground					
	H11	G	Ground					
	H12	G	Ground					
	H13	G	Ground					
	H14	G	Ground					
	H16	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	H17	G	Ground					
	H18	G	Ground					
	H19	G	Ground					
	H20	G	Ground					
	H21	G	Ground					
	H22	G	Ground					
	H24	G	Ground					
	H25	G	Ground					
	H26	G	Ground					
	H29	G	Ground					
	H30	G	Ground					
	H31	G	Ground					
	H32	G	Ground					
	H33	G	Ground					
	H34	G	Ground					
	J3	G	Ground					
	J4	G	Ground					
	J5	G	Ground					
	J6	G	Ground					
	J7	G	Ground					
	J8	G	Ground					
	J9	G	Ground					
	J10	G	Ground					
	J11	G	Ground					
	J22	G	Ground					
	J25	G	Ground					
	J28	G	Ground					
	J29	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	J32	G	Ground					
	K6	G	Ground					
	K8	G	Ground					
	K9	G	Ground					
	K10	G	Ground					
	K17	G	Ground					
	K26	G	Ground					
	K27	G	Ground					
	K30	G	Ground					
	K31	G	Ground					
	K32	G	Ground					
	L1	G	Ground					
	L2	G	Ground					
	L4	G	Ground					
	L5	G	Ground					
	L6	G	Ground					
	L7	G	Ground					
	L8	G	Ground					
	L9	G	Ground					
	L24	G	Ground					
	L26	G	Ground					
	L28	G	Ground					
	L29	G	Ground					
	L32	G	Ground					
	M3	G	Ground					
	M4	G	Ground					
	M5	G	Ground					
	M8	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	M14	G	Ground					
	M16	G	Ground					
	M18	G	Ground					
	M20	G	Ground					
	M22	G	Ground					
	M26	G	Ground					
	M28	G	Ground					
	M30	G	Ground					
	M31	G	Ground					
	N1	G	Ground					
	N2	G	Ground					
	N3	G	Ground					
	N6	G	Ground					
	N7	G	Ground					
	N8	G	Ground					
	N9	G	Ground					
	N10	G	Ground					
	N12	G	Ground					
	N16	G	Ground					
	N18	G	Ground					
	N20	G	Ground					
	N22	G	Ground					
	N24	G	Ground					
	N26	G	Ground					
	N28	G	Ground					
	N30	G	Ground					
	N31	G	Ground					
	P3	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	P4	G	Ground					
	P5	G	Ground					
	P6	G	Ground					
	P9	G	Ground					
	P10	G	Ground					
	P16	G	Ground					
	P18	G	Ground					
	P22	G	Ground					
	P26	G	Ground					
	P27	G	Ground					
	R7	G	Ground					
	R8	G	Ground					
	R14	G	Ground					
	R16	G	Ground					
	R22	G	Ground					
	R26	G	Ground					
	R28	G	Ground					
	T1	G	Ground					
	T9	G	Ground					
	T13	G	Ground					
	T14	G	Ground					
	T16	G	Ground					
	T22	G	Ground					
	T24	G	Ground					
	T30	G	Ground					
	U2	G	Ground					
	U3	G	Ground					
	U7	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	U8	G	Ground					
	U9	G	Ground					
	U11	G	Ground					
	U18	G	Ground					
	U20	G	Ground					
	U22	G	Ground					
	U26	G	Ground					
	U27	G	Ground					
	U28	G	Ground					
	U30	G	Ground					
	V1	G	Ground					
	V9	G	Ground					
	V11	G	Ground					
	V13	G	Ground					
	V16	G	Ground					
	V18	G	Ground					
	V20	G	Ground					
	V22	G	Ground					
	V24	G	Ground					
	V25	G	Ground					
	V27	G	Ground					
	V28	G	Ground					
	V29	G	Ground					
	V30	G	Ground					
	W6	G	Ground					
	W9	G	Ground					
	W11	G	Ground					
	W16	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	W18	G	Ground					
	W22	G	Ground					
	W30	G	Ground					
	Y4	G	Ground					
	Y11	G	Ground					
	Y13	G	Ground					
	Y18	G	Ground					
	Y22	G	Ground					
	Y24	G	Ground					
	Y26	G	Ground					
	Y27	G	Ground					
	Y28	G	Ground					
	Y30	G	Ground					
	AA6	G	Ground					
	AA15	G	Ground					
	AA17	G	Ground					
	AA20	G	Ground					
	AA22	G	Ground					
	AA26	G	Ground					
	AA30	G	Ground					
	AA31	G	Ground					
	AA32	G	Ground					
	AA33	G	Ground					
	AB5	G	Ground					
	AB6	G	Ground					
	AB11	G	Ground					
	AB13	G	Ground					
	AB15	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	AB17	G	Ground					
	AB26	G	Ground					
	AB27	G	Ground					
	AB28	G	Ground					
	AB29	G	Ground					
	AC1	G	Ground					
	AC2	G	Ground					
	AC6	G	Ground					
	AC15	G	Ground					
	AC17	G	Ground					
	AC20	G	Ground					
	AC22	G	Ground					
	AC24	G	Ground					
	AC27	G	Ground					
	AC29	G	Ground					
	AD1	G	Ground					
	AD2	G	Ground					
	AD3	G	Ground					
	AD4	G	Ground					
	AD5	G	Ground					
	AD11	G	Ground					
	AD13	G	Ground					
	AD15	G	Ground					
	AD17	G	Ground					
	AD22	G	Ground					
	AD26	G	Ground					
	AD27	G	Ground					
	AD28	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	AD30	G	Ground					
	AD31	G	Ground					
	AD34	G	Ground					
	AE3	G	Ground					
	AE4	G	Ground					
	AE6	G	Ground					
	AE7	G	Ground					
	AE9	G	Ground					
	AE13	G	Ground					
	AE17	G	Ground					
	AE20	G	Ground					
	AE22	G	Ground					
	AE24	G	Ground					
	AE27	G	Ground					
	AE28	G	Ground					
	AE32	G	Ground					
	AF1	G	Ground					
	AF2	G	Ground					
	AF5	G	Ground					
	AF6	G	Ground					
	AF7	G	Ground					
	AF9	G	Ground					
	AF11	G	Ground					
	AF14	G	Ground					
	AF15	G	Ground					
	AF17	G	Ground					
	AF20	G	Ground					
	AF24	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	AF26	G	Ground					
	AF29	G	Ground					
	AF34	G	Ground					
	AG4	G	Ground					
	AG6	G	Ground					
	AG7	G	Ground					
	AG9	G	Ground					
	AG11	G	Ground					
	AG13	G	Ground					
	AG15	G	Ground					
	AG17	G	Ground					
	AG22	G	Ground					
	AG26	G	Ground					
	AG27	G	Ground					
	AG28	G	Ground					
	AG29	G	Ground					
	AG30	G	Ground					
	AG31	G	Ground					
	AG32	G	Ground					
	AG33	G	Ground					
	AH1	G	Ground					
	AH4	G	Ground					
	AH5	G	Ground					
	AH6	G	Ground					
	AH7	G	Ground					
	AH9	G	Ground					
	AH13	G	Ground					
	AH15	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	AH17	G	Ground					
	AH22	G	Ground					
	AH24	G	Ground					
	AH26	G	Ground					
	AH27	G	Ground					
	AH28	G	Ground					
	AH29	G	Ground					
	AH32	G	Ground					
	AH33	G	Ground					
	AJ1	G	Ground					
	AJ6	G	Ground					
	AJ7	G	Ground					
	AJ9	G	Ground					
	AJ17	G	Ground					
	AJ20	G	Ground					
	AJ22	G	Ground					
	AJ24	G	Ground					
	AJ26	G	Ground					
	AJ27	G	Ground					
	AJ29	G	Ground					
	AJ30	G	Ground					
	AJ31	G	Ground					
	AJ34	G	Ground					
	AK3	G	Ground					
	AK6	G	Ground					
	AK7	G	Ground					
	AK9	G	Ground					
	AK11	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	AK12	G	Ground					
	AK13	G	Ground					
	AK14	G	Ground					
	AK15	G	Ground					
	AK16	G	Ground					
	AK17	G	Ground					
	AK19	G	Ground					
	AK20	G	Ground					
	AK21	G	Ground					
	AK22	G	Ground					
	AK23	G	Ground					
	AK24	G	Ground					
	AK25	G	Ground					
	AK27	G	Ground					
	AK28	G	Ground					
	AK29	G	Ground					
	AK32	G	Ground					
	AK33	G	Ground					
	AL5	G	Ground					
	AL6	G	Ground					
	AL7	G	Ground					
	AL9	G	Ground					
	AL27	G	Ground					
	AL28	G	Ground					
	AL29	G	Ground					
	AL30	G	Ground					
	AL31	G	Ground					
	AL32	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	AL33	G	Ground					
	AM1	G	Ground					
	AM6	G	Ground					
	AM7	G	Ground					
	AM28	G	Ground					
	AM29	G	Ground					
	AM30	G	Ground					
	AM31	G	Ground					
	AM34	G	Ground					
	AN1	G	Ground					
	AN2	G	Ground					
	AN3	G	Ground					
	AN6	G	Ground					
	AN29	G	Ground					
	AN32	G	Ground					
	AN33	G	Ground					
	AP4	G	Ground					
	AP7	G	Ground					
	AP18	G	Ground					
	AP19	G	Ground					
	AP23	G	Ground					
	AP25	G	Ground					
	AP30	G	Ground					
	AP31	G	Ground					
	AP32	G	Ground					
	AP34	G	Ground					
	AR1	G	Ground					
	AR6	G	Ground					

Ball Name	Ball location	Type	Description	Power Domain	PU/PD	I/O Reset Default Mode	Signal Name	Aux Function
	AR7	G	Ground					
	AR8	G	Ground					
	AR30	G	Ground					
	AR31	G	Ground					
	AR32	G	Ground					
	AR34	G	Ground					
	AT4	G	Ground					
	AT5	G	Ground					
	AT6	G	Ground					
	AT32	G	Ground					
	AT33	G	Ground					
Electrostatic discharge (ESD)								
CDM3P5A	J33	G	Reserved					
CDM5P5A	J34	G	Reserved					

4 Electrical Characteristics

4.1 Absolute Maximum Ratings

4.1.1 Free-Air and Storage Thermal Specification

Table 4-1. Absolute maximum ratings for ambient temperature

Parameter	Min.	Max.	Unit
Free-air temperature range	-40	85	°C
Storage temperature range	-55	150	°C

4.1.2 Power Supply

This section lists the maximum values. Note that the values should **not** be exceeded.

Table 4-2. Absolute maximum ratings for power supply

Ball Name	Description	Max.	Unit
AVDD15_WBG	Analog power	1.65	V
AVDD12_WBG	Analog power	1.32	V
AVDD12_UFS	Analog power	1.32	V
AVDD15_UFS	Analog power	1.65	V
AVDD12_CKBUF_UFS	Analog power	1.32	V
AVDD12_USB	Analog power	1.32	V
AVDD15_USB	Analog power	1.65	V
AVDD33_USB	Analog power	3.21	V
AVDD15_PROC	Analog power	1.65	V
AVDD12_MCUPLL	Analog power	1.32	V
AVDD15_MCUPLL	Analog power	1.65	V
AVDD12_NPUPLL	Analog power	1.32	V
AVDD15_NPUPLL	Analog power	1.65	V
AVDD12_GPUPLL	Analog power	1.32	V
AVDD15_GPUPLL	Analog power	1.65	V
AVDD12_MDPLL	Analog power	1.32	V
AVDD15_MDPLL	Analog power	1.65	V
AVDD12_PLL	Analog power	1.32	V
AVDD15_PLL	Analog power	1.65	V
AVDD15_CKSQ	Analog power	1.65	V
AVDD12_CKSQ	Analog power	1.32	V
AVDD12_DRF	Analog power	1.32	V
AVDD15_DRF	Analog power	1.65	V

Ball Name	Description	Max.	Unit
AVDD12_CSI	Analog power	1.32	V
AVDD08_DRV_DSI	Analog power	0.88	V
AVDD12_DSI	Analog power	1.32	V
AVDD15_DSI	Analog power	1.65	V
AVDD12_SSUSB_P1	Analog power	1.32	V
AVDD15_SSUSB_P1	Analog power	1.65	V
AVDD12_PCIE	Analog power	1.32	V
AVDD15_PCIE	Analog power	1.65	V
AVDD12_EMI	Analog DDRPHY power	1.26	V
AVDD15_EMI	Analog DDRPHY power	1.65	V
AVDDQ_EMI0/1	Analog DDRPHY power	0.63	V
AVDD075_EMI0/1	Analog DDRPHY power	0.89	V
DVDD18_IOLM	Digital I/O power	1.98	V
DVDD18_IORT	Digital I/O power	1.98	V
DVDD18_IOBM	Digital I/O power	1.98	V
DVDD18_IORB	Digital I/O power	1.98	V
DVDD12_IOBM_MIPI	Digital I/O power	1.28	V
DVDD18_IOBM_MIPI	Digital I/O power	1.9	V
DVDD12_IORT_HS	Digital I/O power	1.28	V
DVDD18_IORT_HS	Digital I/O power	1.9	V
DVDD15_MSDC1	Digital I/O power	1.28	V
DVDD28_MSDC1	Digital I/O power	1.9	V
DVDD15_SIM	Digital I/O power	1.6	V
DVDD28_SIM1	Digital I/O power	3.15	V
DVDD28_SIM2	Digital I/O power	3.15	V
DVDD12_ESIM	Digital I/O power	1.28	V
DVDD18_ESIM	Digital I/O power	1.9	V
DVDD18_VQPS	Digital power	1.98	V
DVDD_PROC_L	Digital power	0.9	V
DVDD_PROC_B	Digital power	1.05	V
DVDD_SRAM_PROC_L	Digital power	0.9	V
DVDD_SRAM_PROC_B	Digital power	1.05	V
DVDD_NPU	Digital power	0.85	V
DVDD_SRAM_NPU	Digital power	0.85	V
DVDD_GPUSTACK	Digital power	0.85	V
DVDD_SRAM_GPUSTACK	Digital power	0.85	V
DVDD_MM	Digital power	0.7	V
DVDD_SRAM_MM	Digital power	0.75	V
DVDD_MODEM	Digital power	0.825	V
DVDD_SRAM_MODEM	Digital power	0.825	V

Ball Name	Description	Max.	Unit
DVDD_SRAM_CORE	Digital power	0.825	V
DVDD_CORE	Digital power	0.825	V

4.2 Recommended Operating Conditions

This section lists the recommended conditions for device operations.

4.2.1 Thermal Operating Specification

Table 4-3. Thermal operating specification

Symbol	Description	Min.	Max. ¹	Unit
T _{ambient} T _{junction}	Operating junction temperature	T _{ambient} = -40	T _{junction} = 125	°C

(1) Full electrical specifications are guaranteed only up to 105°C junction temperature.

4.2.2 Power Supply

Table 4-4. Recommended operating conditions for power supply

Ball Name	Description	Min.	Typ.	Max.	Unit
AVDD08_DRV_DSI	Analog power	0.7125	0.75	0.84	V
AVDD12_WBG	Analog power	1.14	1.2	1.26	V
AVDD12_UFS	Analog power	1.14	1.2	1.26	V
AVDD12_USB	Analog power	1.14	1.2	1.26	V
AVDD12_CKBUF_UFS	Analog power	1.14	1.2	1.26	V
AVDD12_CKSQ	Analog power	1.14	1.2	1.26	V
AVDD12_DRF	Analog power	1.14	1.2	1.26	V
AVDD12_CSI	Analog power	1.14	1.2	1.26	V
AVDD12_DSI	Analog power	1.14	1.2	1.26	V
AVDD12_MCUPLL	Analog power	1.14	1.2	1.26	V
AVDD12_NPUPLL	Analog power	1.14	1.2	1.26	V
AVDD12_GPUPLL	Analog power	1.14	1.2	1.26	V
AVDD12_MDPLL	Analog power	1.14	1.2	1.26	V
AVDD12_PLL	Analog power	1.14	1.2	1.26	V
AVDD12_SSUSB_P1	Analog power	1.14	1.2	1.26	V
AVDD12_PCIE	Analog power	1.14	1.2	1.26	V
AVDD15_UFS	Analog power	1.425	1.5	1.575	V
AVDD15_WBG	Analog power	1.425	1.5	1.575	V
AVDD15_USB	Analog power	1.425	1.5	1.575	V
AVDD15_DSI	Analog power	1.425	1.5	1.575	V
AVDD15_CKSQ	Analog power	1.425	1.5	1.575	V
AVDD15_DRF	Analog power	1.425	1.5	1.575	V
AVDD15_SSUSB_P1	Analog power	1.425	1.5	1.575	V
AVDD15_PCIE	Analog power	1.425	1.5	1.575	V

Ball Name	Description	Min.	Typ.	Max.	Unit
AVDD15_MCUPLL	Analog power	1.425	1.5	1.575	V
AVDD15_NPUPLL	Analog power	1.425	1.5	1.575	V
AVDD15_GPUPLL	Analog power	1.425	1.5	1.575	V
AVDD15_MDPLL	Analog power	1.425	1.5	1.575	V
AVDD15_PROC	Analog power	1.425	1.5	1.575	V
AVDD15_PLL	Analog power	1.425	1.5	1.575	V
AVDD33_USB	Analog power	2.85	3	3.15	V
AVDDQ_EMI0/1	Analog DDRPHY power	0.57	0.6	0.63	V
AVDD075_EMI0/1	Analog DDRPHY power	0.55	0.75	0.7875	V
AVDD12_EMI	Analog DDRPHY power	1.14	1.2	1.26	V
AVDD15_EMI	Analog DDRPHY power	1.425	1.5	1.575	V
DVDD_PROC_L	Digital power	0.81	0.9	0.99	V
		0.675	0.75	0.825	V
		0.495	0.55	0.605	V
DVDD_PROC_B	Digital power	0.945	1.05	1.155	V
		0.81	0.9	0.99	V
		0.675	0.75	0.825	V
		0.495	0.55	0.605	V
DVDD_SRAM_PROC_L	Digital power	0.81	0.9	0.99	V
		0.675	0.75	0.825	V
DVDD_SRAM_PROC_B	Digital power	0.945	1.05	1.155	V
		0.81	0.9	0.99	V
		0.675	0.75	0.825	V
DVDD_NPU	Digital power	0.81	0.9	0.99	V
		0.675	0.75	0.825	V
		0.585	0.65	0.715	V
		0.495	0.55	0.605	V
DVDD_SRAM_NPU	Digital power	0.81	0.9	0.99	V
		0.675	0.75	0.825	V
DVDD_GPUSTACK	Digital power	0.765	0.85	0.935	V
		0.608	0.675	0.743	V
		0.495	0.55	0.605	V
DVDD_SRAM_GPUSTACK	Digital power	0.765	0.85	0.935	V
		0.675	0.75	0.825	V
DVDD_MM	Digital power	0.63	0.7	0.77	V
		0.585	0.65	0.715	V
		0.54	0.6	0.66	V
		0.518	0.575	0.632	V
DVDD_SRAM_MM	Digital power	0.675	0.75	0.825	V
DVDD_MODEM	Digital power	0.72	0.8	0.88	V

Ball Name	Description	Min.	Typ.	Max.	Unit
		0.675	0.75	0.825	V
		0.608	0.675	0.742	V
		0.563	0.625	0.687	V
		0.518	0.575	0.632	V
		0.495	0.55	0.605	V
DVDD_SRAM_MODEM	Digital power	0.72	0.8	0.88	V
		0.675	0.75	0.825	V
		0.495	0.55	0.605	V
DVDD_SRAM_CORE	Digital power	0.743	0.825	0.907	V
		0.675	0.75	0.825	V
		0.495	0.55	0.605	V
DVDD_CORE	Digital power	0.743	0.825	0.907	V
		0.653	0.725	0.797	V
		0.585	0.65	0.715	V
		0.54	0.6	0.66	V
		0.518	0.575	0.632	V
		0.495	0.55	0.605	V
DVDD18_VQPS	Digital power	1.62	1.8	1.98	V
DVDD18_IOLM	Digital I/O power	1.62	1.8	1.98	V
DVDD18_IORT	Digital I/O power	1.62	1.8	1.98	V
DVDD18_IOBM	Digital I/O power	1.62	1.8	1.98	V
DVDD18_IORB	Digital I/O power	1.62	1.8	1.98	V
DVDD12_IOBM_MIPI	Digital I/O power	1.12	1.2	1.28	V
DVDD18_IOBM_MIPI	Digital I/O power	1.7	1.8	1.9	V
DVDD12_IORT_HS	Digital I/O power	1.12	1.2	1.28	V
DVDD18_IORT_HS	Digital I/O power	1.7	1.8	1.9	V
DVDD15_MSDC1	Digital I/O power	1.12	1.2	1.28	V
DVDD28_MSDC1	Digital I/O power	1.7	1.8	1.9	V
DVDD15_SIM	Digital I/O power	1.4	1.5	1.6	V
DVDD28_SIM1	Digital I/O power	2.7	3	3.15	V
DVDD28_SIM2	Digital I/O power	2.7	3	3.15	V
DVDD12_ESIM	Digital I/O power	1.12	1.2	1.28	V
DVDD18_ESIM	Digital I/O power	1.7	1.8	1.9	V

4.3 DC Electrical Characteristics

This section provides DC parameters of the device.

4.3.1 X32K_IN DC Electrical Characteristics

Table 4-5. X32K_IN DC electrical characteristics (DVDD18_IOBM = 1.8V)

Parameter	Description	Min.	Typ.	Max.	Unit
V _{IH}	High-level input voltage	0.65*DVDD18_IOBM	-	DVDD18_IOBM + 0.3	V
V _{IL}	Low-level input voltage	-0.3	-	0.35*DVDD18_IOBM	V
V _{OH}	High-level output voltage	0.75*DVDD18_IOBM	-	-	V
V _{OL}	Low-level output voltage	-	-	0.25*DVDD18_IOBM	V
Fx32k_in	Input clock frequency	-	32	-	kHz
DCx32k_in	Input signal duty cycle	45	50	55	%

4.3.2 SPI DC Electrical Characteristics

Table 4-6. SPI DC electrical characteristics (DVDD18_IOxx_xx = 1.8V)

Parameter	Description	Min.	Typ.	Max.	Unit
V _{IH}	High-level input voltage	0.65* DVDD18_IOxx_xx	-	DVDD18_IOxx_xx + 0.3	V
V _{IL}	Low-level input voltage	-0.3	-	0.35* DVDD18_IOxx_xx	V
V _{OH}	High-level output voltage	0.75* DVDD18_IOxx_xx	-	-	V
V _{OL}	Low-level output voltage	-	-	0.25* DVDD18_IOxx_xx	V

4.3.3 I2S DC Electrical Characteristics

Table 4-7. I2S DC electrical characteristics (DVDD18_IOxx_xx = 1.8V)

Parameter	Description	Min.	Typ.	Max.	Unit
V _{IH}	High-level input voltage	0.65* DVDD18_IOxx_xx	-	DVDD18_IOxx_xx + 0.3	V
V _{IL}	Low-level input voltage	-0.3	-	0.35* DVDD18_IOxx_xx	V
V _{OH}	High-level output voltage	0.75* DVDD18_IOxx_xx	-	-	V
V _{OL}	Low-level output voltage	-	-	0.2* DVDD18_IOxx_xx	V

4.3.4 I2C/I3C DC Electrical Characteristics

Table 4-8. I2C/I3C DC electrical characteristics (DVDD18_IOxx_xx = 1.8V)

Parameter	Description	Min.	Typ.	Max.	Unit
V _{IH}	High-level input voltage	0.65* DVDD18_IOxx_xx	-	DVDD18_IOxx_xx + 0.3	V
V _{IL}	Low-level input voltage	-0.3	-	0.35* DVDD18_IOxx_xx	V
V _{OL}	Low-level output voltage	-	-	0.2* DVDD18_IOxx_xx	V

4.3.5 MSDC DC Electrical Characteristics

Table 4-9. MSDC1 DC electrical characteristics (DVDD28_MSDC1 = 3V)

Parameter	Description	Min.	Typ.	Max.	Unit
V _{IH}	Input logic high voltage	0.625* DVDD28_MSDC1	-	DVDD28_MSDC1+0.3	V
V _{IL}	Input logic low voltage	-0.3	-	0.25* DVDD28_MSDC1	V
V _{OH}	DC output logic high voltage	0.75* DVDD28_MSDC1	-	DVDD28_MSDC1+0.3	V
V _{OL}	DC output logic low voltage	-0.3	-	0.125* DVDD28_MSDC1	V

Table 4-10. MSDC1 DC electrical characteristics (DVDD28_MSDC1 = 1.8V)

Parameter	Description	Min.	Typ.	Max.	Unit
V _{IH}	Input logic high voltage	0.7* DVDD28_MSDC1	-	DVDD28_MSDC1+0.3	V
V _{IL}	Input logic low voltage	-0.3	-	0.3* DVDD28_MSDC1	V
V _{OH}	DC output logic high voltage	1.4	-	DVDD28_MSDC1+0.3	V
V _{OL}	DC output logic low voltage	-0.3	-	0.45	V

4.3.6 SIM DC Electrical Characteristics

Table 4-11. SIM DC electrical characteristics

Parameter	Condition	Symbol	Min.	Typ.	Max.	Unit
SIM1_SIO						
High-level input voltage	DVDD28_SIM1 = 1.8V	V _{ih}	1.35	1.8	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.45	V
High-level output voltage		V _{oh}	1.53	1.8	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.27	V
High-level input voltage	DVDD28_SIM1 = 3.0V	V _{ih}	1.875	3.0	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.75	V
High-level output voltage		V _{oh}	2.55	3.0	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.45	V
SIM1_SCLK						
High-level input voltage	DVDD28_SIM1 = 1.8V	V _{ih}	1.35	1.8	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.45	V
High-level output voltage		V _{oh}	1.53	1.8	N/A	V

Parameter	Condition	Symbol	Min.	Typ.	Max.	Unit
Low-level output voltage	DVDD28_SIM1 = 3.0V	V _{ol}	N/A	0.0	0.27	V
High-level input voltage		V _{ih}	1.875	3.0	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.75	V
High-level output voltage		V _{oh}	2.55	3.0	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.45	V
SIM1_SRST						
High-level input voltage	DVDD28_SIM1 = 1.8V	V _{ih}	1.35	1.8	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.45	V
High-level output voltage		V _{oh}	1.53	1.8	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.27	V
High-level input voltage	DVDD28_SIM1 = 3.0V	V _{ih}	1.875	3.0	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.75	V
High-level output voltage		V _{oh}	2.55	3.0	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.45	V
SIM2_SIO						
High-level input voltage	DVDD28_SIM2 = 1.8V	V _{ih}	1.35	1.8	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.45	V
High-level output voltage		V _{oh}	1.53	1.8	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.27	V
High-level input voltage	DVDD28_SIM2 = 3.0V	V _{ih}	1.875	3.0	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.75	V
High-level output voltage		V _{oh}	2.55	3.0	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.45	V
SIM2_SCLK						
High-level input voltage	DVDD28_SIM2 = 1.8V	V _{ih}	1.35	1.8	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.45	V
High-level output voltage		V _{oh}	1.53	1.8	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.27	V
High-level input voltage	DVDD28_SIM2 = 3.0V	V _{ih}	1.875	3.0	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.75	V
High-level output voltage		V _{oh}	2.55	3.0	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.45	V
SIM2_SRST						
High-level input voltage	DVDD28_SIM2 = 1.8V	V _{ih}	1.35	1.8	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.45	V
High-level output voltage		V _{oh}	1.53	1.8	N/A	V
Low-level output voltage		V _{ol}	N/A	0.0	0.27	V
High-level input voltage	DVDD28_SIM2 = 3.0V	V _{ih}	1.875	3.0	N/A	V
Low-level input voltage		V _{il}	N/A	0.0	0.75	V
High-level output voltage		V _{oh}	2.55	3.0	N/A	V

Parameter	Condition	Symbol	Min.	Typ.	Max.	Unit
Low-level output voltage		V_{ol}	N/A	0.0	0.45	V

4.4 AC Electrical Characteristics and Timing Diagrams

4.4.1 External Memory Interface for LPDDR5

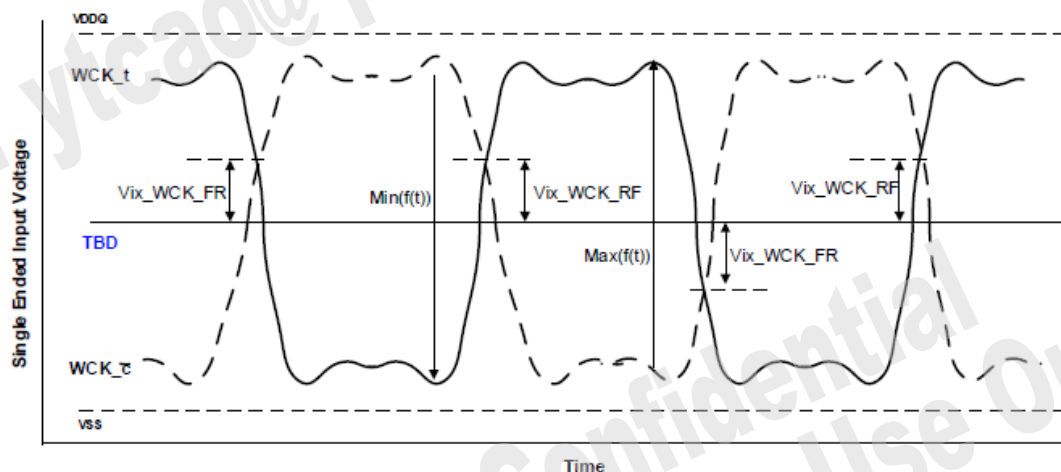


Figure 4-1. LPDDR5 WCK VIX definition

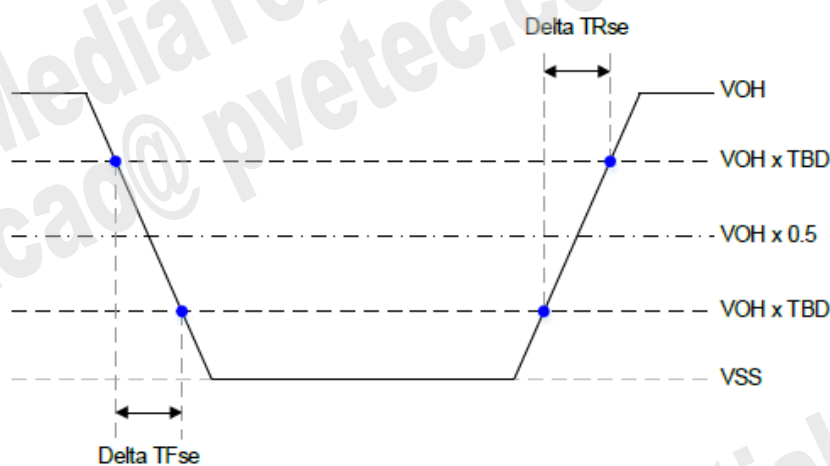


Figure 4-2. LPDDR5 single-ended output slew-rate definition

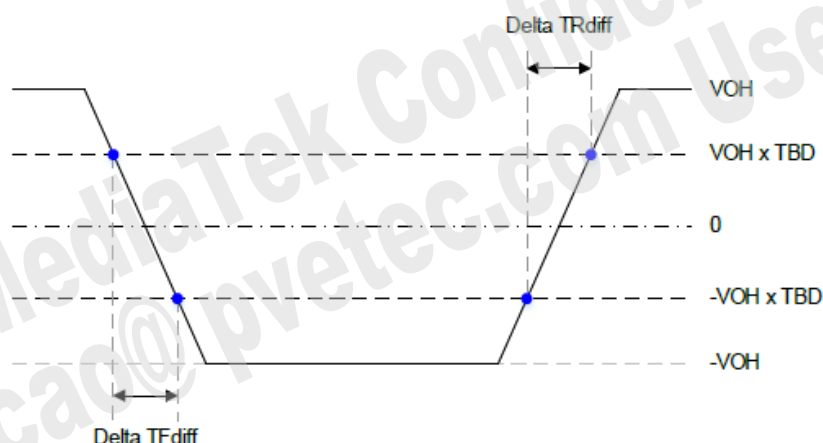


Figure 4-3. LPDDR5 differential output slew-rate definition

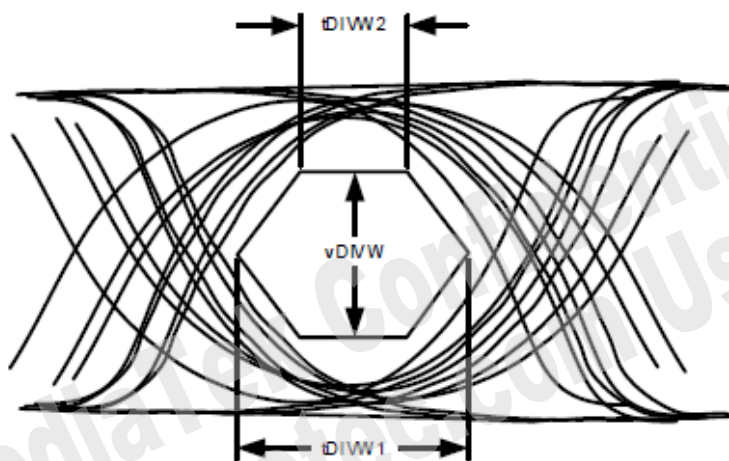


Figure 4-4. LPDDR5 RX mask

Table 4-12. LPDDR5 AC timing parameter table of external memory interface

Symbol	Description	Min.	Typ.	Max.	Unit
Vix_WCK_ratio	WCK differential input crosspoint voltage ratio	-	-	20	%
Vix_CK_ratio	CK differential input crosspoint voltage ratio	-	-	25	%
vDIVW	DQ RX mask height	-	-	100	mV
tDIVW1	DQ RX mask width	-	-	0.35	UI
tDIVW2	DQ RX mask width at vDIVW	-	-	0.18	UI

4.4.2 SPI AC Timing Characteristics

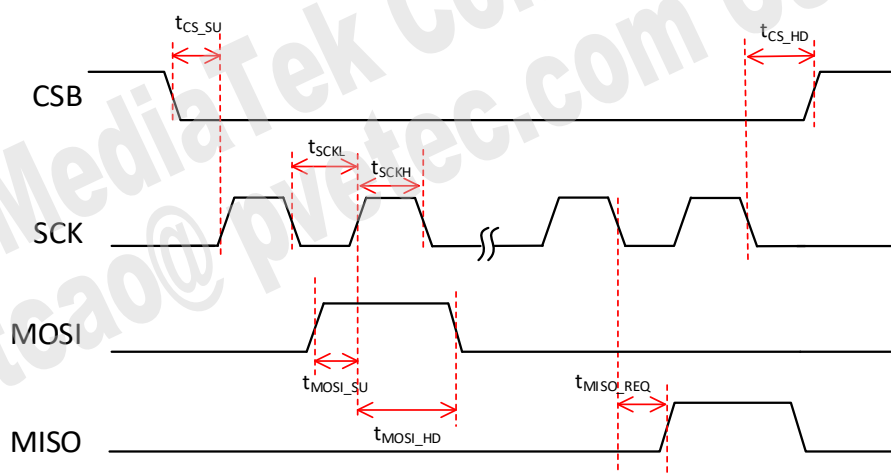


Figure 4-5. SPI timing diagram

Table 4-13. SPI AC timing parameters

Symbol	Description	Performance			Unit
		Min.	Typ.	Max.	
f_{SCK}	SPI master SCK clock frequency	-	-	52	MHz
t_{MOSI_SU}	MOSI to SCK rising setup time	6	-	-	ns
t_{MOSI_HD}	SCK rising to MOSI hold time	6	-	-	ns
t_{SCKL}	SCK low pulse	7.2	-	-	ns
t_{SCKH}	SCK high pulse	7.2	-	-	ns
t_{CSB_SU}	CSB falling to SCK rising setup time	1.8	-	-	ns
t_{CSB_HD}	SCK falling to CSB rising hold time	1.8	-	-	ns
t_{MISO_REQ}	SCK falling to MISO change time requirement	0	-	10	ns

Note:

- In the CS GPIO mode, SPI_CS is handled by the software.
 - The software should pull down the SPI_CS pin before the SPI starts transferring.
 - The software should pull up the SPI_CS pin when the SPI completes the transaction.

Based on the sequence above, the minimum specification of t_{CSB_SU} and t_{CSB_HD} time can be satisfied.

- To achieve the max. frequency of SCK, the internal sample clock delay of the SPI master should be adjusted.
- MISO data valid time should be one cycle of f_{SCK} .
- For dual mode, all the output data pins can refer to the MOSI timing parameters, and all the input data pins can refer to the MISO timing parameters.

4.4.3 I2S AC Timing Characteristics

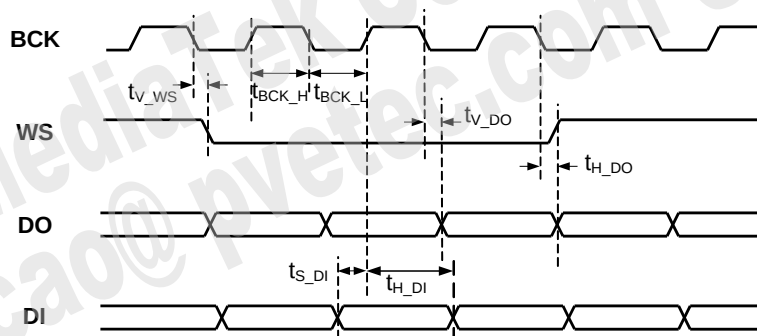


Figure 4-6. I2S master mode timing diagram

Table 4-14. I2S AC timing parameters

Parameter	Description	Min.	Typ.	Max.	Unit
f_s	Sampling frequency	8	-	192	kHz
t_{ws}	Word select period	32	-	64	1/fBCK
f_{MCK}	Master clock frequency	-	-	24.576	MHz
f_{BCK}	Serial clock frequency	$32 \cdot f_s$	-	$64 \cdot f_s$	MHz
t_{BCK_H}	BCK high-level time	-	0.5	-	1/fBCK
t_{BCK_L}	BCK low-level time	-	0.5	-	1/fBCK
t_{V_WS}	WS valid time	-	-	0.2	1/fBCK
t_{H_WS}	WS hold time	0	-	-	1/fBCK
t_{V_DO}	DO valid time	-	-	0.2	1/fBCK
t_{H_DO}	DO hold time	0	-	-	1/fBCK
t_{S_DI}	DI setup time	0.2	-	-	1/fBCK
t_{H_DI}	DI hold time	0.2	-	-	1/fBCK

4.4.4 I2C AC Timing Characteristics

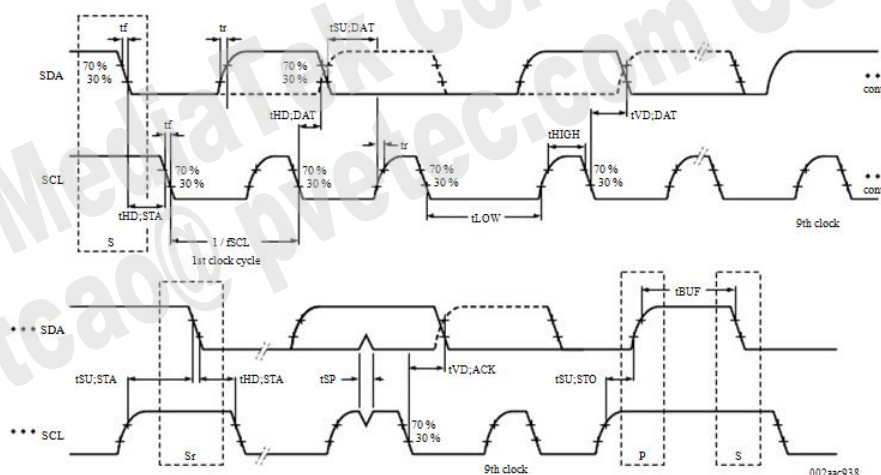


Figure 4-7. I2C AC timing diagram of F/S mode

Table 4-15. I2C AC timing parameters for standard, fast, and fast mode plus

Symbol	Parameter	Standard Mode		Fast Mode		Fast Mode Plus		Unit	Remark
		Min.	Max.	Min.	Max.	Min.	Max.		
f_{SCL}	Serial clock line (SCL) clock frequency	0	100	0	400	0	1,000	kHz	-
$t_{HD,STA}$	Hold time (repeated) START condition	4.0	-	0.6	-	0.26	-	μs	-
t_{LOW}	LOW period of the SCL clock	4.7	-	1.3	-	0.5	-	μs	-
t_{HIGH}	HIGH period of the SCL clock	4.0	-	0.6	-	0.26	-	μs	-
$t_{SU,STA}$	Set-up time for a repeated START condition	4.7	-	0.6	-	0.26	-	μs	-
$t_{HD,DAT}$	Data hold time	5.0	-	0	-	0	-	μs	I2C-bus devices
$t_{SU,DAT}$	Data set-up time	250	-	100	-	50	-	ns	-
t_r	Rise time of both SDA and SCL signals	-	1,000	20	300	-	120	ns	-
t_f	Fall time of both SDA and SCL signals	-	300	20x (VDD/5.5V)	300	20x (VDD/5.5V)	120	ns	VDD is I2C I/O voltage.
$t_{SU,STO}$	Set-up time for STOP condition	4.0	-	0.6	-	0.26	-	ns	-
$t_{VD,DAT}$	Data valid time	-	3.45	-	0.9	-	0.45	μs	-
$t_{VD,ACK}$	Data valid acknowledge time	-	3.45	-	0.9	-	0.45	μs	-

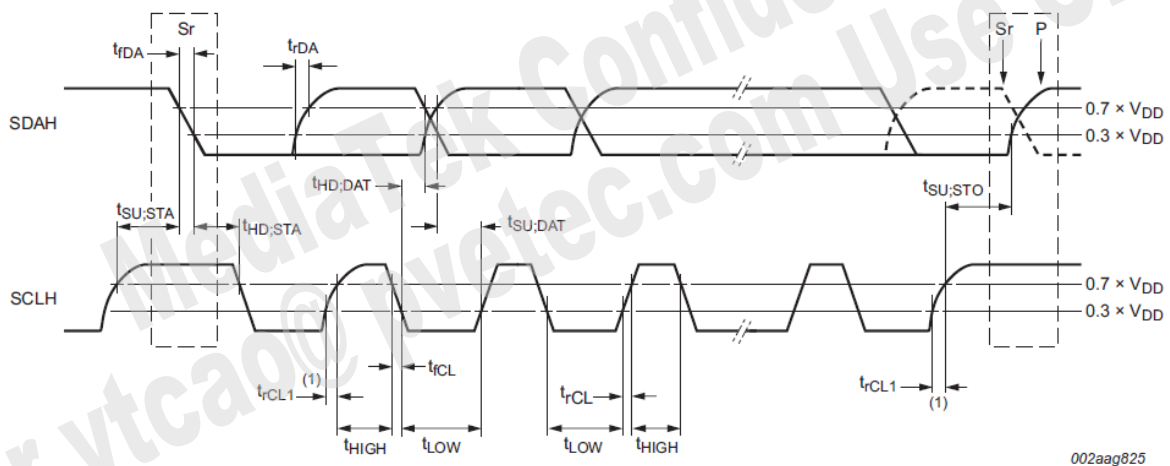


Figure 4-8. I2C AC timing diagram of HS mode

Table 4-16. I2C AC timing parameters for HS mode

Symbol	Parameter	Cb = 100 pF (max.)		Cb = 400 pF		Unit	Note
		Min.	Max.	Min.	Max.		
f _{SCL}	SCL clock frequency	0	3.4	0	1.7	MHz	-
t _{SU,STA}	Set-up time (repeated) START condition	160	-	160	-	ns	-
t _{HD,STA}	Hold time (repeated) START condition	160	-	160	-	ns	-
t _{LOW}	LOW period of the SCL clock	160	-	320	-	ns	-
t _{HIGH}	HIGH period of the SCL clock	60	-	120	-	ns	-
t _{HD,DAT}	Data hold time	0	70	0	150	ns	I2C-bus devices
t _{SU,DAT}	Data set-up time	10	-	10	-	ns	-
t _r	Rise time of SCLH signal	10	40	20	80	ns	-
t _f	Fall time of SCLH signal	10	40	20	80	ns	-
t _{SU,STO}	Set-up time for STOP condition	160	-	160	-	ns	-

4.4.5 MSDC AC Timing Characteristics

4.4.5.1 Default Speed Timing

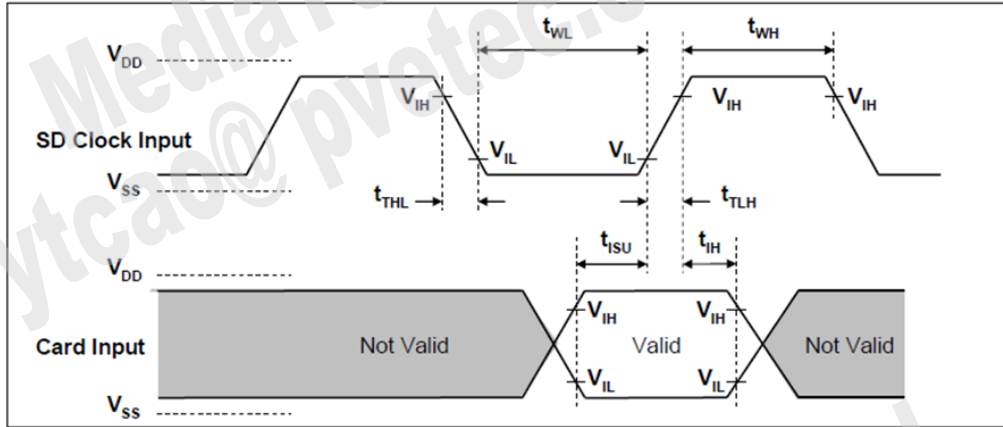


Figure 4-9. MSDC device input timing diagram of default speed

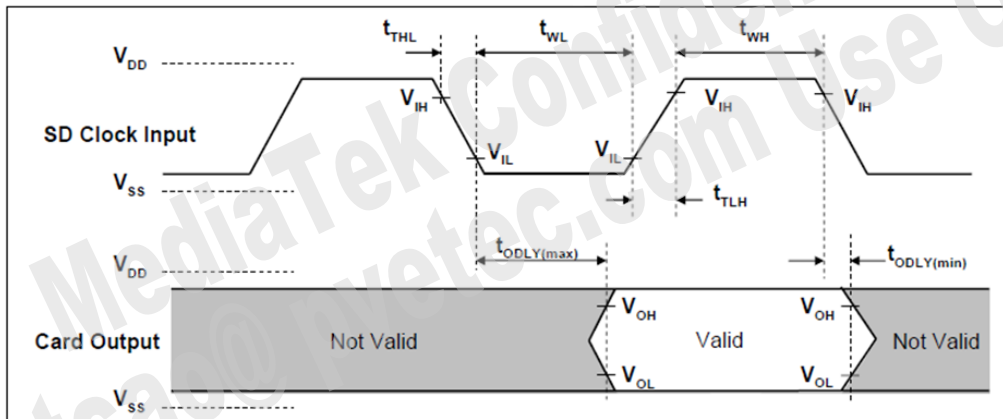


Figure 4-10. MSDC device input timing diagram of default speed

Table 4-17. MSDC device AC timing parameters of default speed

Parameter	Symbol	Min.	Max.	Unit
Clock CLK (All values are referred to min. (V_{IH}) and max. (V_{IL}))				
Clock frequency data transfer mode	f_{PP}	0	25	MHz
Clock frequency identification mode	f_{OD}	0/100	400	kHz
Clock low time	t_{WL}	10	-	ns
Clock high time	t_{WH}	10	-	ns
Clock rise time	t_{TLH}	-	10	ns
Clock fall time	t_{THL}	-	10	ns
Input CMD, DAT (referenced to CLK)				
Input setup time	T_{ISU}	5	-	ns
Input hold time	T_{IH}	5	-	ns
Output CMD, DAT (referenced to CLK)				

Parameter	Symbol	Min.	Max.	Unit
Output delay time during data transfer mode	T_{ODLY}	0	14	ns
Output delay time during identification mode	T_{ODLY}	0	50	ns

4.4.5.2 High Speed Timing

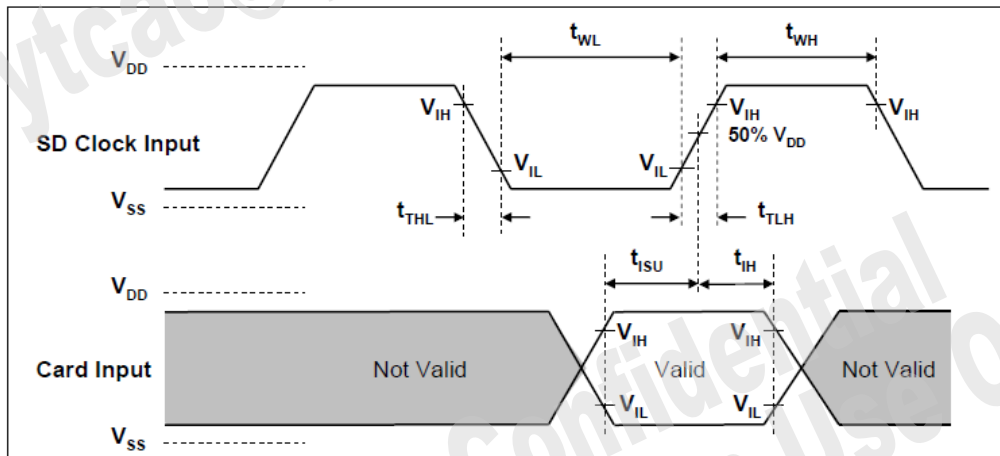


Figure 4-11. MSDC device input timing diagram of high speed

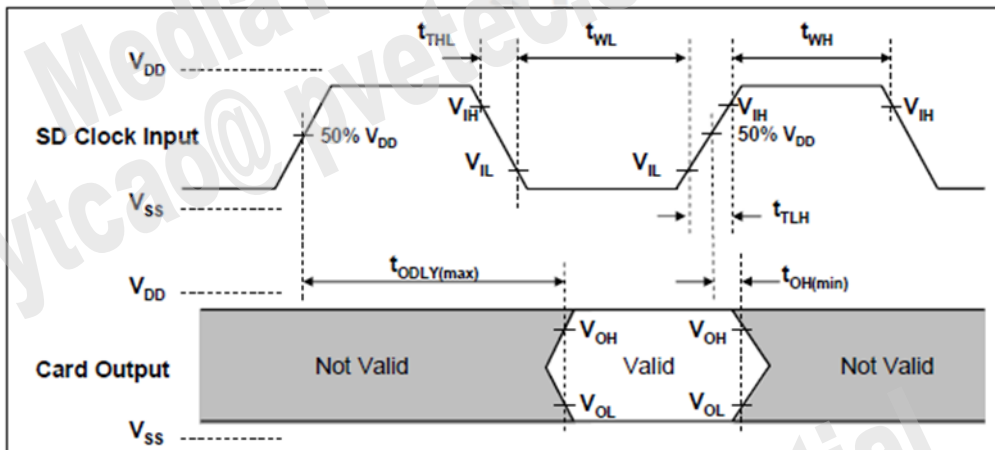


Figure 4-12. MSDC device output timing diagram of high speed

Table 4-18. MSDC device AC timing parameters of high speed

Parameter	Symbol	Min.	Max.	Unit
Clock CLK (All values are referred to min. (V_{IH}) and max. (V_{IL}))				
Clock frequency data transfer mode	f_{PP}	0	50	MHz
Clock low time	t_{WL}	7	-	ns
CLK high time	t_{WH}	7	-	ns
CLK rise time	t_{TLH}	-	3	ns

Parameter	Symbol	Min.	Max.	Unit
CLK fall time	t_{THL}	-	3	ns
Input CMD, DAT (referenced to CLK)				
Input setup time	t_{ISU}	6	-	ns
Input hold time	t_{IH}	2	-	ns
Output CMD, DAT (referenced to CLK)				
Output delay time during data transfer mode	t_{ODLY}	-	14	ns
Output hold time	t_{OH}	2.5	-	ns
Total system capacitance for each line	C_L	-	40	pF

4.4.5.3 SDR12/SDR25/SDR50/SDR104 Mode Timing

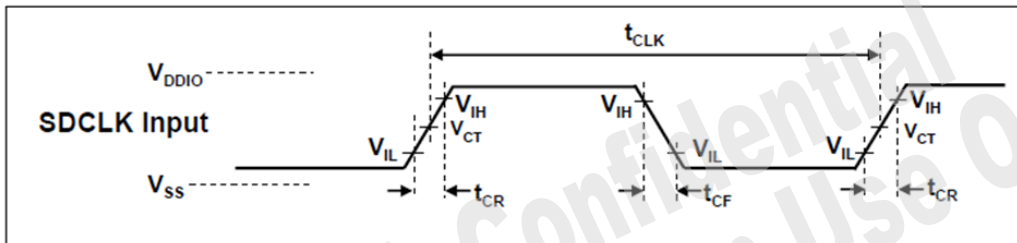


Figure 4-13. MSDC device clock timing diagram of SDR12/SDR25/SDR50/SDR104 mode

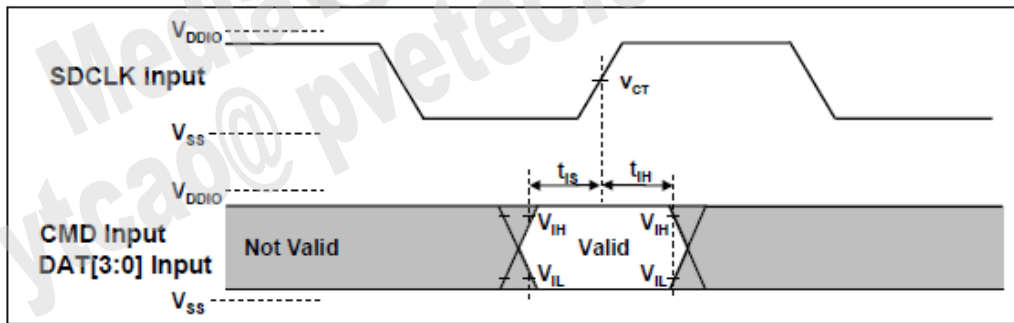


Figure 4-14. MSDC device input timing diagram of SDR50/SDR104 mode

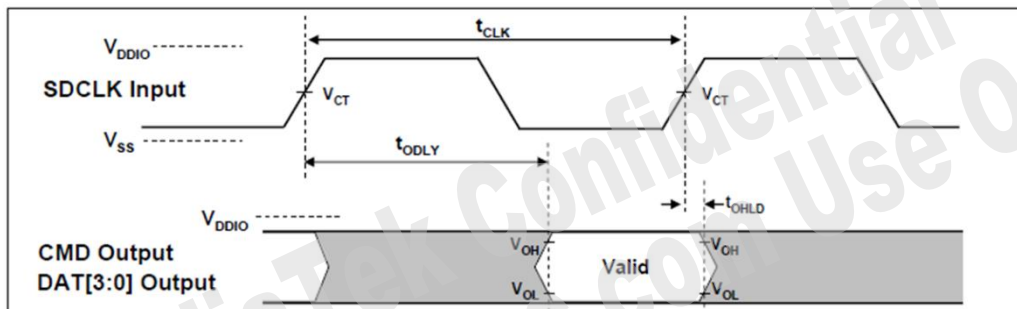


Figure 4-15. MSDC device output timing diagram of fixed data window (SDR12/SDR25/SDR50)

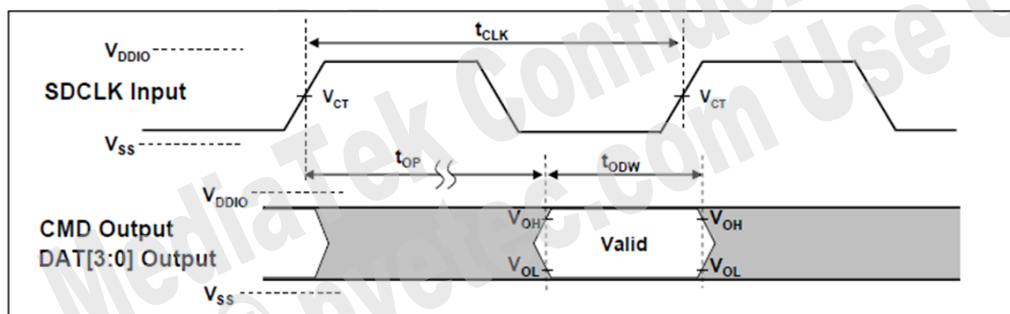


Figure 4-16. MSDC device output timing diagram of variable window (SDR104)

Table 4-19. MSDC device AC timing parameters of SDR12/SDR25/SDR50/SDR104 mode

Symbol	Min.	Max.	Unit	Remark
Clock CLK				
t_{CLK}	4.8	-	ns	208 MHz (max.), between rising edges, $V_{CT} = 0.975V$
t_{CR}, t_{CF}	-	$0.2 \cdot t_{CLK}$	ns	$t_{CR}, t_{CF} < 0.96$ ns (max.) at 208 MHz, $C_{CARD} = 10$ pF $t_{CR}, t_{CF} < 2.00$ ns (max.) at 100 MHz, $C_{CARD} = 10$ pF The absolute maximum value of t_{CR}, t_{CF} is 10 ns regardless of clock frequency.
Clock duty	30	70	%	-
Input CMD, DAT (SDR104)				
t_{IS}	1.40	-	ns	$C_{CARD} = 10$ pF, $V_{CT} = 0.975V$
t_{IH}	0.80	-	ns	$C_{CARD} = 5$ pF, $V_{CT} = 0.975V$
Input CMD, DAT (SDR50)				
t_{IS}	3.00	-	ns	$C_{CARD} = 10$ pF, $V_{CT} = 0.975V$
t_{IH}	0.80	-	ns	$C_{CARD} = 5$ pF, $V_{CT} = 0.975V$
Output CMD, DAT (SDR12/SDR25/SDR50)				
t_{ODLY}	-	7.5	ns	$t_{CLK} \geq 10.0$ ns, $C_L = 30$ pF, using driver type B, for SDR50
t_{ODLY}	-	14	ns	$t_{CLK} \geq 20.0$ ns, $C_L = 40$ pF, using driver type B, for SDR25 and SDR12
T_{OH}	1.5	-	ns	Hold time at the t_{ODLY} (min.), $C_L = 15$ pF
Output CMD, DAT (SDR104)				
t_{OP}	0	2	UI	Card output phase
Δt_{OP}	-350	+1,550	ps	Delay variation due to temperature change after tuning.
t_{ODW}	0.6	-	UI	$t_{ODW} = 2.88$ ns at 208 MHz

4.4.5.4 DDR50 Speed Mode Timing

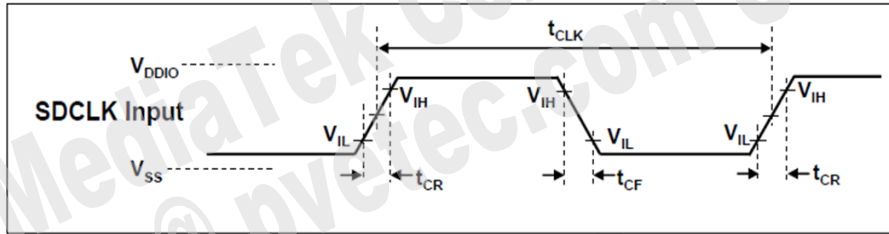


Figure 4-17. MSDC device clock timing diagram of DDR50 speed mode

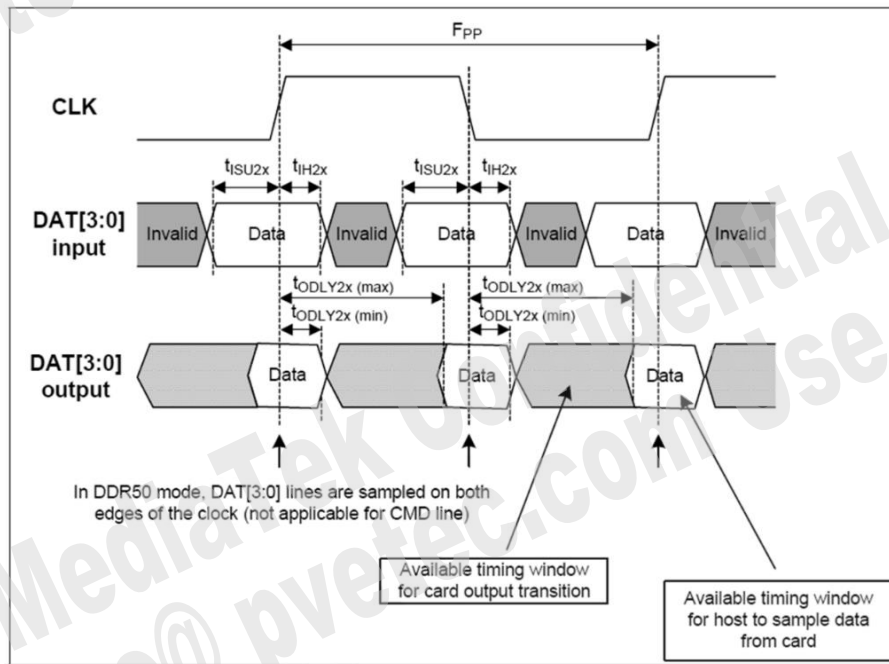


Figure 4-18. MSDC device input/output timing diagram of DDR50 speed mode

Table 4-20. MSDC device AC timing parameters of DDR50 speed mode

Parameter	Symbol	Min.	Max.	Unit
Clock CLK				
Clock period	t_{CLK}	20	-	ns
Clock rise time	t_{CR}, t_{CF}	-	$0.2 * t_{CLK}$	ns
Clock fall time				
Clock duty	-	45	55	%
Input CMD (referenced to CLK rising edge)				
Input setup time	t_{ISU}	6	-	ns
Input hold time	t_{IH}	0.8	-	ns
Output CMD (referenced to CLK rising edge)				
Output delay time during data transfer mode	t_{ODLY}	-	13.7	ns
Output hold time	t_{OH}	1.5	-	ns
Input DAT (referenced to CLK rising and falling edges)				

Parameter	Symbol	Min.	Max.	Unit
Input setup time	t_{ISU2x}	3	-	ns
Input hold time	t_{IH2x}	0.8	-	ns
Output DAT (referenced to CLK rising and falling edges)				
Output delay time during data transfer mode	t_{ODLY2x}	-	7.0	ns
Output hold time	t_{OH2x}	1.5	-	ns

4.4.6 SIM AC Timing Characteristics

Table 4-21. SIM AC timing parameters

Parameter	Condition	Symbol	Min.	Typ.	Max.	Unit
SIM1_SCLK						
Rise and fall time	DVDD28_SIM1 = 1.8V	T _{rise_fall}	N/A	50	50	ns
Clock duty		Duty	47	50	53	%
Rise and fall time	DVDD28_SIM1 = 3.0V	T _{rise_fall}	N/A	18	18	ns
Clock duty		Duty	47	50	53	%
SIM1_SIO						
Rise and fall time	DVDD28_SIM1 = 1.8V	T _{rise_fall}	N/A	50	1,000	ns
Rise and fall time	DVDD28_SIM1 = 3.0V	T _{rise_fall}	N/A	18	1,000	ns
SIM1_SRST						
Rise and fall time	DVDD28_SIM1 = 1.8V	T _{rise_fall}	N/A	50	1,000	ns
Rise and fall time	DVDD28_SIM1 = 3.0V	T _{rise_fall}	N/A	18	1,000	ns
SIM2_SCLK						
Rise and fall time	DVDD28_SIM2 = 1.8V	T _{rise_fall}	N/A	50	50	ns
Clock duty		Duty	47	50	53	%
Rise and fall time	DVDD28_SIM2 = 3.0V	T _{rise_fall}	N/A	18	18	ns
Clock duty		Duty	47	50	53	%
SIM2_SIO						
Rise and fall time	DVDD28_SIM2 = 1.8V	T _{rise_fall}	N/A	50	1,000	ns
Rise and fall time	DVDD28_SIM2 = 3.0V	T _{rise_fall}	N/A	18	1,000	ns
SIM2_SRST						
Rise and fall time	DVDD28_SIM2 = 1.8V	T _{rise_fall}	N/A	50	1,000	ns
Rise and fall time	DVDD28_SIM2 = 3.0V	T _{rise_fall}	N/A	18	1,000	ns

4.5 Clock Characteristics

The clock squarer is designed to receive clock signal from pin “X26M_IN” and distribute it to the chip internally. See Table 4-22 for the functional specifications of clock squarer of pin “X26M_IN”.

Table 4-22. Clock squarer specifications

Symbol	Parameter	Min.	Typ.	Max.	Unit
V _{in}	Input signal amplitude	1,000	1,200	1,250	mVpp
D _{cyc} IN	Input signal duty cycle	-	50		%
D _{cyc} OUT	Output signal duty cycle	D _{cyc} IN - 5	-	D _{cyc} IN + 5	%
	Maximum positive overshoot	-	-	1.3	V
	Minimum negative overshoot	-0.1	-	-	V

5 Power On Sequence

The power on sequence described in this chapter is the power predesigned to operate with other different MediaTek chipsets. For details, refer to the following power management IC (PMIC) documents.

- *MT6363 PMIC Design Notice for MT8668*
- *MT6373 PMIC Design Notice for MT8668*

Furthermore, for the power on sequence allowing user-control, refer to MT8668 Application Processor Datasheet.

6 Boot Mode Configuration

Boot mode configuration refers to the process of specifying the manner in which a device starts up, accesses system resources and launches the operating system. There are several mode choices to determine the method with which the system locates, loads and executes the boot loader.

6.1 Mode Selection

Table 6-1. MT8668 mode selection

Pin Name	Description	Parameter
KPCOLO	Force download mode	0: Force USB download mode in bootrom 1: Disable (default)
[0] SCP_VREQ_VAO [1] AUD_SYNC_MOSI	JTAG mode	00: AP MCU JTAG (default) 01: AP MCU JTAG + IO JTAG + DAP JTAG 10: AP MCU JTAG + IO JTAG 11: AP MCU JTAG + WCN JTAG
[0] AUD_DAT_MOSIO [1] AUD_DAT_MOSI1 [2] AUD_DAT_MOSI2 [3] EINT_CHG_IRQB	DDR selection mode	0010: LP5 MCP 0011: LP5 discrete (default)

6.2 Constant Tie Pins

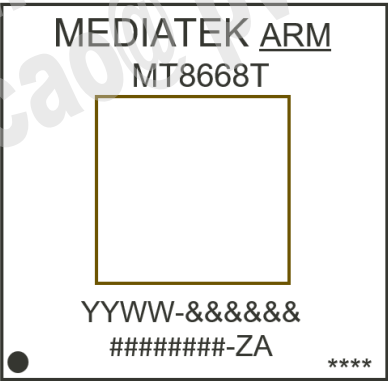
Table 6-2. MT8668 constant tie pin

Pin Name	Description
TESTMODE	Test mode (tied to GND)

7 Package Information

7.1 Top Marking

MT8668T/**ZA**



MT8668T: Part number
YYWW: Date code
&&&&&: Random code
#####: Die lot number
ZA: Function code
****: Trace code

Figure 7-1. MT8668 top marking

7.2 Ordering Information

Table 7-1. MT8668 ordering information

Order#	Marking	Temperature Range ⁽¹⁾	Package
MT8668	See Section 7.1.	-40°C ~ 85°C	TFBGA

(1) The temperature range refers to the IC ambient temperatures.

7.3 Package Outlines

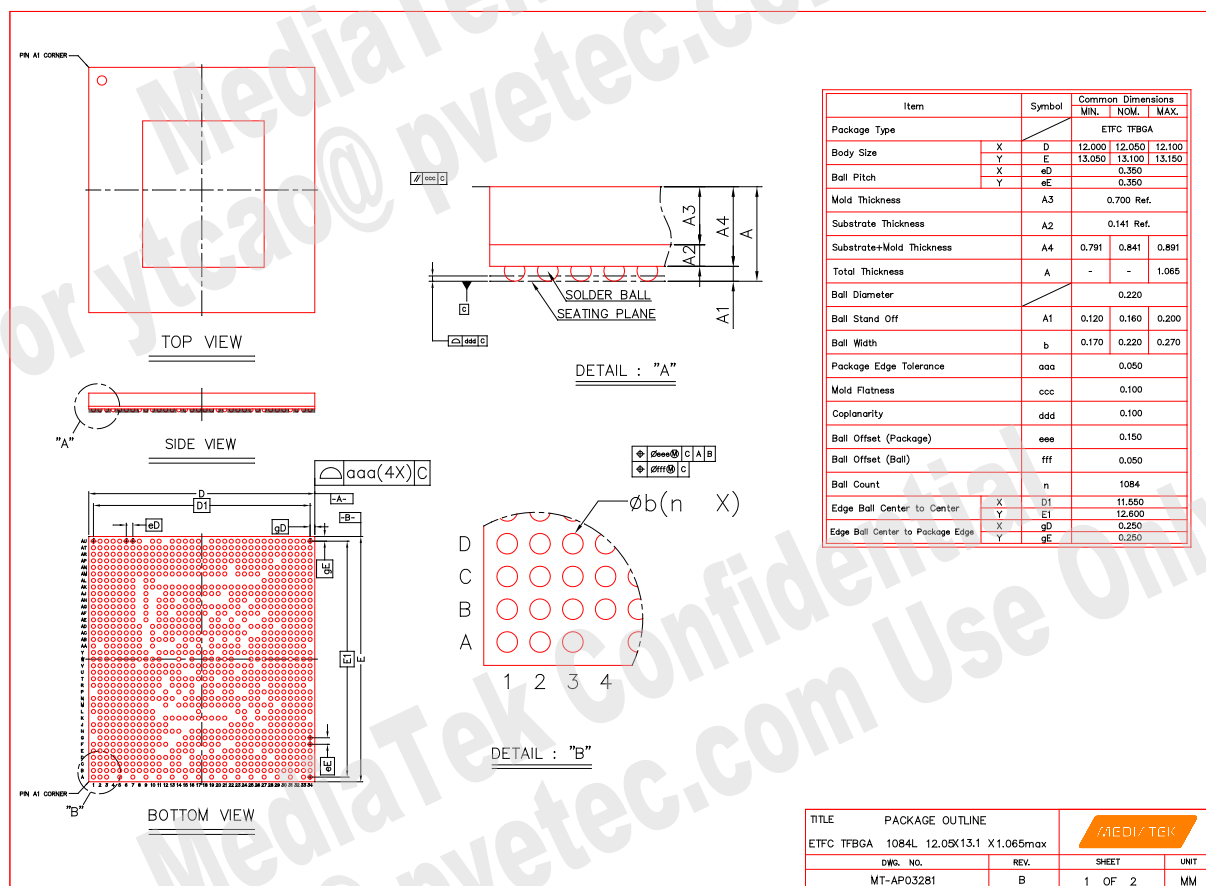


Figure 7-2. Outlines and dimensions of TFBGA 12.05 mm x 13.1 mm, 1084-ball, 0.35 mm pitch package

8 Reference

8.1 Reference Documents

Module	Reference
Power on sequence	
PMIC	MT6363_Design_Notice_for_MT8668
	MT6373_Design_Notice_for_MT8668

8.2 MT8668 Companion Chips

The capabilities of the companion chips listed below are for reference only. For the exact details and operating conditions, please refer to the corresponding datasheet or design document in accordance with the part name.

Table 8-1. MT8668 companion chips

Part Name	Category	Description	Reference Document
MT6373	2nd PMIC	- Input range: 2.6V ~ 5.0V 10 buck converters and 25 LDOs optimized for specific subsystems - SPMI interface - Over-current and thermal overload protection - Programmable under voltage lockout protection - Watchdog reset - Flexibility hardware PMIC reset function - Power-on reset and start-up timer - Storage card plug-out protection 166-pin WFBGA package	MT6373_PMIC_Datasheet
MT6197	RF TX/RX	- Full multi-mode RF solution (GGE/C2K/WCDMA/LTE/NR SA/NR NSA) - Multi-band NR sub6G/LTE/WCDMA/C2K/GCE 256QAM UL/256QAM DL (HSPA+/LTE) - C2K/2G/3G/4G/5G co-banding - Supports RxD. - Direct conversion transmitter (NR/LTE/3G/8-PSK) and DFM for 2G GMSK - Dedicated power detection circuits for power control over specific power range 3 LMHB + 1 CB (NR-U) + 2 all-band TX0 output ports 3 LMHB + 1 CB (NR-U) + 2 all-band TX1 output ports - TX0 and TX1 can be configured as CA/ENDC or 2x2 MIMO.	MT6197_RF_datasheet_1p4_Release

Part Name	Category	Description	Reference Document
		- Hybrid direct-conversion (5G/4G/3G/C2K)/Low-IF (GGE, DC-HSDPA) receiver 20 RX input ports - Digital output interface - External LNA at RxP and RxD are required. - Supports 2G DRX & 3G/4G/5G TDD SAWless RX. - Low supply current and operation directly from DC-DC converters - Supports RF calibration features for key RX and TX specifications (image rejection, LO feed-through, IIP2, DC offset, RC corner). - Temperature measurement subsystem - LTE carrier aggregation - Supports 2UL + 3DL inter-band CA. - Max. CCA BW up to 100 MHz - Supports RX 4x4 MIMO for LB/MHB/CB including n46 and n96. - Supports TX 2x2 MIMO for LMHB and CB (NR-U). - NR carrier aggregation - Supports 2UL + 2DL inter-band CA. - Max. CCA BW up to 200 MHz - LTE + NR dual connectivity support 2UL + 3DL inter-band ENDC. - Supports NSA/SA+LTE DR-DSDS. - Supports power Class 2 high-power UE (HPUE) in HB/CB.	
MT6685	CLK IC	- Handles 5G-module baseband clock IC - Input range: 2.6V ~ 5V 9 clock buffers and 4 input channels for ADC optimized for specific 5G-module subsystems 32K-crystal-less RTC oscillator for system timing, 1.8V clock buffer output - SPMI interface - Programmable under voltage lockout protection - Watchdog reset - Flexibility hardware clock IC reset function - Power-on reset and start-up timer 42-pin WFBGA package	MT6685_DCXO_Datasheet
MT6363	Main PMIC	10 buck converters and 26 LDOs - SPMI interface - Over-current and thermal overload protection - Programmable under voltage lockout protection - Watchdog reset - Flexibility hardware PMIC reset function - Power-on reset and start-up timer - Storage card plug-out protection	MT6363_PMIC_Datasheet

Part Name	Category	Description	Reference Document
MT6637W	CONSYS RF	167-pin WFBGA package - MT6637X is a 4-in-1 connectivity chip which contains two 2.4 GHz Wi-Fi transceiver front-ends, two 5 GHz or 6 GHz Wi-Fi transceiver front-ends, a Bluetooth transceiver front-end, a GPS L1-band receiver front-end, a GPS L5-band receiver front-end and a complete FM receiver in a WFBGA package. - Supports tri-band 2.4 GHz/5 GHz/6 GHz. - Supports worldwide Wi-Fi 5G/6G channel, and Bluetooth FDD operation. - Supports 1*1 Wi-Fi 5/6 GHz and Wi-Fi 2.4 GHz dual-band dual-concurrency. - Supports external PA and LNA for Wi-Fi-2.4 GHz and Wi-Fi-5 GHz/6 GHz. - RF supports simultaneous dual-band L1 and L5 operation. - The system feature depends on companion modem chip's capability.	MT6637X_External_DataSheet
MT6308H	PMIC	- Buck power stage to provide low-frequency power to PA. - Class-AB amplifier to provide high-frequency modulation of PA supply voltage. - Buck and amplifier power combined using ac-coupling capacitor (external). - Linked control system to implement power split. - Buck regulator to provide optimal power supply for class-AB amplifier. - APT mode operation in which buck regulator produces a constant-voltage output. - Ability to transit from envelope tracking (ET) mode to average power tracking (APT) mode on the fly. - Adaptive pass-device sizing to optimize efficiency of buck converters. - MIPI RFFE serial interface to baseband. -3 dB bandwidth up to 90 MHz. - VHFP output voltage up to 507V with 6.0W power delivery.	MT6308H_DataSheet

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